



Bahir Dar University

Bahir Dar Institute of Technology

Faculty of Computing

Requirement Analysis Document (RAD)

for

Industrial Project On Digital Notice Board System for Bahir Dar Institute of Technology

(BIT)

**Submitted to the faculty of computing in partial fulfillment of the requirements for the
degree of Bachelor of Information System**

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2018

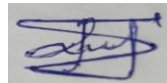
Bahir Dar University, Bahir Dar Institute of Technology

Declaration

The Project is our own and has not been presented for a degree in any other university and all the sources of material used for the project have been duly acknowledged.

We hereby declare that this project titled "Digital Notice Board System for BIT" is our original work and has not been submitted for any other degree or examination at this or any other institution. All sources of information have been duly acknowledged.

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Persistent and Class modeling	✓	✓
Component and Deployment modeling	✓	✓

ACKNOWLEDGEMENT

We first and foremost thank God Almighty for giving us strength and wisdom to accomplish this project.

We would like to extend our sincerest appreciation to the Information System Department at Bahir Dar Institute of Technology for giving us the opportunity for this project within the framework of our senior year program. We are particularly appreciative for the contributions made by our advisor, Ms.Mahlet W. whose advice and support have been invaluable for this project.

Our appreciation goes to the Faculty of Computing and all our lecturers for providing us with the necessary knowledge to engage in this work. To name a few, we would like to thank the administration of BIT, including the Registrar's Office and Student Services Directorate, for cooperating with us in gathering requirements for this work.

We also thank the students and staff of the campuse who contributed to our survey and interview results that are essential to system development.

Lastly, we would like to thank our family and friends for their support and inspiration during our academic pursuit.

List of acronyms

List of acronyms	Description
BIT	Bahir Dar Institute of Technology
RAD	Requirement Analysis Document
UML	Unified Modeling Language
UI/UX	User Interface/User Experience
E-R	Entity-Relationship
FCM	Firebase Cloud Messaging
API	Application Programming Interface
DBMS	Database Management System
HTTP	Hypertext Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure
JSON	JavaScript Object Notation
MVC	Model View Controller
CRUD	Create, Read, Update, Delete
OOAD	Object Oriented Analysis & Design
PDF	Portable Document Format
SMS	Short Message Service
BSc	Bachelor of Science
ICT	Information Communication Technology
LMS	Learning Management System
WCAG	Web Content Accessibility Guidelines
REST	Representational State Transfer
PWA	Progressive Web Application
RPO	Recovery Point Objective

RTO	Recovery Time Objective
CDN	Content Delivery Network
OS	Operating System
CPU	Central Processing Unit
RAM	Random Access Memory
SSD	Solid State Drive
GUI	Graphical User Interface
ERD	Entity Relationship Diagram
SSO	Single Sign-On
MFA	Multi-Factor Authentication
DB	Database
AI	Artificial Intelligence
IoT	Internet of Things
TLS	Transport Layer Security
VPN	Virtual Private Network
QA	Quality Assurance

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ABSTRACT

This project presents the requirement analysis and design of a Digital Notice Board System for Bahir Dar Institute of Technology (BiT), a major institute within Bahir Dar University. The system aims to address communication challenges associated with traditional paper-based notice boards, such as delayed information dissemination, limited accessibility, and excessive paper usage.

To identify system requirements, data were collected through interviews, questionnaires, and document analysis involving selected stakeholders within the institute. Based on the analysis, functional and non-functional requirements were defined to guide the system design. The proposed solution is a web-based application that allows authorized department staff to publish notices digitally, while students can access notices in a timely and convenient manner.

The system design is modeled using UML diagrams, including use case and sequence diagrams, and follows Object-Oriented Analysis and Design principles. Emphasis is placed on usability, security, and scalability. The proposed Digital Notice Board System is expected to improve communication efficiency within Bahir Dar Institute of Technology and can be extended to other departments or institutes in the future.

Chapter One: Introduction

1.1 Background

Bahir Dar University is one of the biggest public universities in Ethiopia and was established on May 6, 2000, through the merging of Bahir Dar Polytechnic and Bahir Dar Teachers' College. The university consists of several institutes, colleges, faculties, academies, and schools.

Bahir Dar Institute of Technology (BiT) is one of the major institutes within Bahir Dar University and is among the oldest and most recognized technology institutes in Ethiopia. The institute was originally established in 1963 through a bilateral agreement between the Government of Ethiopia and the former Soviet Union as Bahir Dar Polytechnic Institute. Over time, the institute expanded its academic programs in various technology fields, including engineering and applied sciences, and was officially upgraded to Bahir Dar Institute of Technology in 2011.

BiT plays a significant role in producing skilled graduates who contribute to national development in industrial, educational, agricultural, and technological sectors. Currently, the institute enrolls thousands of undergraduate and postgraduate students across multiple engineering and technology programs and is supported by well-equipped laboratories, qualified academic staff, and strong research and community service activities.

Due to its large student population and diverse academic programs, effective and timely communication is essential within the institute. However, traditional paper-based notice boards are limited in efficiency and accessibility. Therefore, there is a need for a Digital Notice Board System that can enhance information dissemination and support modern communication practices within Bahir Dar Institute of Technology

1.2 Statement of the problem

At Bahir Dar Institute of Technology (BiT), the dissemination of academic and administrative information is primarily carried out using traditional paper-based notice boards. Although this method has been used for many years, it presents several challenges that limit effective communication within the institute.

One major issue is the limited physical space available on notice boards, which often leads to the removal of important notices to accommodate new ones. The manual process of preparing, printing, and posting notices also causes delays in information dissemination, preventing students and staff from receiving timely updates. In addition, paper-based notices are not searchable, making it difficult for students to access past announcements related to academic schedules, examinations, or administrative matters.

Accessibility is another concern, as physical notice boards may not be easily accessible to all students, including those with physical disabilities or those who are not present on campus at the time notices are posted. Furthermore, paper notices are vulnerable to damage from weather

conditions such as rain and sunlight, as well as vandalism or unauthorized removal, resulting in loss of important information.

The current system also lacks the ability to target notices to specific departments or student groups and does not support real-time communication for urgent announcements. These limitations reduce the effectiveness of information sharing and create communication gaps within the institute. Therefore, there is a need for a Digital Notice Board System that can provide timely, reliable, and accessible communication for Bahir Dar Institute of Technology.

1.3 Objectives

1.3.1 General Objective

The general objective of this project is to design and develop a Digital Notice Board System for Bahir Dar Institute of Technology (BiT) that improves the dissemination of academic and administrative information by replacing the existing manual, paper-based notice board system with a web-based digital solution.

1.3.2 Specific Objectives

These are the specific objectives which provide a roadmap for the development and implementation of the Digital Notice Board System:

- Assess the existing notice dissemination process and identify communication challenges within Bahir Dar Institute of Technology through observation and stakeholder consultation.
- Design and develop a web-based administrative interface that allows authorized department staff to create, manage, and publish digital notices efficiently.
- Develop a centralized digital system for storing and managing notices and user information in a secure database.
- Enable students to access notices easily through an online platform with basic filtering and viewing functionalities.
- Ensure the system is user-friendly, secure, and scalable to allow future expansion to other departments or institutes.
- Test the system to verify that it meets functional and non-functional requirements and performs reliably within the selected institute.

1.4 Methodology

1.4.1 Requirement gathering methods

Under the criterion for methodology, we will address the way in which the requirements of the users have been identified. The aim is to ensure the requirements of the software are identified. Below are the methodologies used:

1.Interviews:

Structured interviews were conducted with selected stakeholders, including department heads, administrative staff, and student representatives within the institute. These interviews helped to collect qualitative information about the challenges faced when using traditional notice boards and the expectations from a digital system.

2.Document Analysis:

Relevant communication-related documents such as notice formats, academic announcements, and internal communication guidelines were reviewed. In addition, existing physical notice boards and past notices were examined to identify gaps and inefficiencies in the current system.

3 Observation:

Direct observation was carried out to study how students and staff interact with physical notice boards within the institute. This method helped to understand notice accessibility, frequency of updates, and common problems such as overcrowding, outdated notices, and missed information.

4. Literature:

Relevant literature, including academic papers and case studies on digital notice board systems and educational information systems, was reviewed. This helped to identify best practices and guided the design of the proposed system.

1.4.2 Analysis and design Methodology

Analysis and Design Approach

The Object-Oriented (OO) analysis and design method has been selected based on the following considerations:

- It enables scalability and reuse, which is a requirement for the proposed system.

- Provides functionality to graphically represent the system using UML diagrams like use case diagram, class diagram, and sequence diagram in order to effectively communicate the system to different campuses.
- OO Principles promote encapsulation, thereby enabling the effective and proper handling of confidential information such as user information as well as administrative correspondences.

Tools for Analysis and Design

1. Project Management

- **ClickUp:** Used for planning, task management, and tracking project progress throughout the development process.

2. System and UI Design

- **Figma:** Used for designing user interfaces, wireframes, and basic prototypes of the system.
- **Draw.io:** Used for creating UML diagrams such as use case and sequence diagrams.

1.4.3 Implementation Methodology

The platform's implementation was guided by the following tools and technologies to ensure efficient development, debugging, and maintenance:

1. Programming Language:

- **JavaScript:** Selected for its versatility, robust ecosystem, and ability to handle both frontend and backend development with consistent syntax.

2. Database Management System (DBMS):

- **MySQL:** A reliable relational database system that efficiently manages structured data for BIT's notice board requirements, user information, and attachments.

3. Application Framework:

- **React.js:** Enables building a responsive, component-based web application with efficient state management for BIT's user interface.
- **Express.js:** Provides a minimal, flexible Node.js web application framework for building robust backend APIs and routing.

4. Styling Framework:

- **CSS3 with custom styling:** Utilizes modern CSS features including Flexbox and Grid for responsive design, ensuring the interface works across different devices used at BIT.

5. Authentication and Security:

- **JWT (JSON Web Tokens):** Implements secure session management for user authentication.
- **bcrypt:** Provides password hashing to ensure user credential security.
- **BIT email verification:** Ensures only BIT community members can register using institutional emails.

6. File Upload System:

- **Multer:** Middleware for handling multipart/form-data, primarily used for uploading notice attachments.

7. API Testing Tool:

- **Postman:** Used for testing and validating RESTful API endpoints during development.

8. Version Control:

- **Git with GitHub:** Facilitates collaborative development, version control, and efficient code management.

9. Development Environment:

- **Visual Studio Code:** Provides an efficient, customizable coding environment with extensive extensions for JavaScript development.

1.5 Feasibility

The viability of the project is also considered to establish whether it is possible to implement. This will include evaluating economic gains of the project, level of technical expertise required, as well as time needed to completely implement it in seven college campuses.

Economic Feasibility

It is economically feasible because of the following reasons:

- **Cost Efficiency:** Utilization of open-source technologies including React.js, Node.js, and MySQL results in minimal development costs for BIT.

- **Long-term Savings:** Elimination of paper usage, printing expenses, and physical board maintenance will generate operational savings for BIT. Currently, the institute incurs significant expenses in printing and distributing notices.

- **Value Addition:** The system enhances communication efficiency, which may translate to improved academic and administrative performance within BIT.

- **Infrastructure Costs are Minimal:** Cloud hosting solutions work to ensure that there are minimal costs incurred when it comes to the underlying infrastructure, as everything is hosted on the cloud.

Technical Feasibility

The technical feasibility is justified by:

- **Team Expertise:** The development team possesses skills in JavaScript, React.js, Node.js, and MySQL which are essential for building the system for BIT.
- **Tool Selection:** Selected tools (React.js, Express.js, MySQL) are appropriate for developing a functional system suitable for BIT's requirements.
- **Resource Availability:** Required development tools and resources are readily available and accessible.
- **Compatibility:** The chosen technologies are compatible with BIT's existing infrastructure and commonly used devices.

Time Feasibility

The project timeline has been carefully planned for timely delivery:

- **Detailed Schedule:** Activities have been segmented into manageable stages with defined milestones for execution at BIT.
- **Agile Development:** Agile methodology enables flexibility, rapid iterations, and progress tracking throughout development.
- **Resource Allocation:** Adequate resource allocation ensures tasks can be completed within the allocated timeframe.
- **Buffer Time:** Contingency time has been included to accommodate unforeseen delays during development.

1.6 Beneficiaries or significant of the project

The proposed system will deliver benefits to various stakeholders within Bahir Dar Institute of Technology, enhancing communication efficiency, information accessibility, and environmental sustainability.

1. For BIT Students

Benefits:

- **Convenient Information Access:** Students can access notifications and announcements anytime, anywhere via internet-connected devices.
- **Time Efficiency:** Students save time previously spent visiting physical notice boards, allowing more focus on academic activities.
- **Timely Updates:** Important updates are delivered immediately, ensuring students never miss deadlines or relevant events.
- **Personalized Information:** Notices can be filtered by department and relevance, reducing information overload.

Significance:

- Improves academic performance through better access to essential information.
- Reduces stress and anxiety caused by missed announcements.
- Enhances digital literacy among students.

2. For BIT Administrative Staff and Faculty

Benefits:

- **Efficient Notice Management:** Staff can create, update, and manage notices digitally without physical limitations.
- **Targeted Communication:** Announcements can be targeted to specific departments or user groups.
- **Reduced Administrative Work:** Automation reduces manual work and minimizes communication errors.
- **Basic Analytics:** Simple view tracking helps assess communication effectiveness.

Significance:

- Enhances operational efficiency and reduces administrative workload.
- Improves quality and timeliness of institutional communications within BIT.

- Provides basic insights for improving communication strategies.

3. For BIT Administration

Benefits:

- **Cost Reduction:** Savings on paper usage, printing, and notice board maintenance.
- **Environmental Sustainability:** Reduced paper consumption aligns with environmental initiatives.
- **Standardized Communication:** Ensures consistent information dissemination across BIT.
- **Digital Transformation:** Positions BIT as a technology-embracing institution.

Significance:

- Demonstrates commitment to digital innovation and operational excellence.
- Enhances institutional reputation through modern communication systems.
- Contributes to BIT's sustainability goals.

4. For Platform Developers (Project Team)

Benefits:

- **Practical Skill Development:** Opportunity to apply software engineering principles to solve real-world communication challenges at BIT.
- **Portfolio Enhancement:** Development of a complete project demonstrating full-stack capabilities.
- **Problem-Solving Experience:** Addressing real institutional needs within a manageable scope.

Significance:

- Prepares students for professional careers in software development.
- Demonstrates practical application of academic knowledge.
- Contributes to local technology ecosystem by developing solutions for educational institutions.

Broader Impact on Educational Institutions

- **Efficiency Model:** Establishes a reference for digitizing communication in Ethiopian higher education institutions.
- **Accessibility Improvement:** Digital access helps bridge information gaps for all students.
- **Innovation Encouragement:** Promotes adoption of digital tools in academic operations.
- **Scalable Foundation:** Creates a model that can be adapted by other departments or institutions.

Contribution to the Academic Domain

This project addresses communication inefficiencies at BIT by replacing traditional physical notice boards with a digital solution. It demonstrates how technology can make academic communication more efficient, accessible, and environmentally sustainable. By focusing on a manageable scope (BIT rather than entire BDU), the project provides a practical model for digital transformation that can be realistically implemented and serve as a foundation for broader institutional adoption.

1.7 Limitations of the Project

The purpose of this project is to create a Digital Notice Board System to replace paper-based communication at **Bahir Dar Institute of Technology (BIT)**. The proposed project addresses accessibility and efficiency issues faced by BIT students and staff through a web-based notice management system with an easy-to-use interface.

Platform Capabilities for BIT:

- An admin portal for creating and managing notices for BIT departments
- Notice distribution targeted to specific departments within BIT
- Email notifications for important announcements
- Basic search and filtering functions for retrieving notices
- Role-based access control for students, faculty, and administrators
- Responsive web design accessible on various devices

Issues Encountered: Challenges and Constraints

1. **Internet Connectivity Dependence:** The system requires internet access, which may be unreliable in some areas. However, as a web application, it remains accessible through campus Wi-Fi and mobile data.
2. **Digital Literacy Gap:** Some faculty members accustomed to traditional methods may need initial training to adopt the digital platform.

3. **Resource Constraints:** As a student project, development is limited by academic timelines and team size.
4. **Initial Feature Set:** Due to time constraints, some advanced features will be implemented as basic versions initially.

Features for Future Enhancement:

- Mobile application development (current focus is web-only)
- Advanced analytics and reporting
- Integration with existing BIT systems (student portal, LMS)
- SMS notifications for urgent alerts
- Offline functionality for limited connectivity scenarios
- Enhanced accessibility features for users with disabilities

1.8 Scope of the Project

Project Focus: The Digital Notice Board System will transform communication processes specifically within **Bahir Dar Institute of Technology (BIT)**.

System Includes:

1. User Management:

- Registration and authentication using BIT email addresses
- Three user roles: Student, Faculty/Staff, Administrator
- Profile management for BIT community members

2. Notice Management:

- Create, edit, and delete notices with text content
- File attachment support (PDF, DOCX, images) up to 5MB
- Notice categorization (Academic, Event, General, Exam)
- Basic scheduling and expiration settings

3. Notice Viewing and Search:

- Chronological display of notices
- Filtering by category and date
- Basic keyword search functionality
- Notice details view with attachments

4. Notification System:

- Email notifications for new notices (optional user preference)
- In-application notification center

5. **Administration Features:**

- User management for administrators
- System settings configuration
- Basic usage statistics

6. **Role-Based Access Control:** Various levels of privileges to be granted to admin, staff, faculty, and student groups with proper restrictions.

7. **Notice Management Capabilities:** Fully featured capabilities for notice creation, notice editing, notice organization, notice scheduling, and notice archiving with attachment support and rich notice content.

8. **User Notification System:** Various options such as alerts, email notifications, and SMS notifications based on important announcements.

9. **Search and Filtering Capabilities:** The ability to search and filter by department, type, date range, and key phrases.

10. **Analytics Dashboard:** Simple reporting functionality for monitoring views, user engagement, and system use trends.

11. **Responsive Design:** This ensures the interface is optimized for use on a desktop computer or a mobile device or even a tablet computer.

Business Processes/Subsystems to be Implemented

1. **User Registration and Authentication**

- **Service Offering:** Secure user registration, login, and management for BIT students, faculty, and staff.
- **Features:**
 - Registration using BIT institutional email verification
 - Basic authentication with email/password
 - Password recovery via email
 - Three user roles: Student, Faculty/Staff, Admin
 - Basic profile management (name, email, department)

2. **Notice Management System**

- **Service Offered:** Toolkit for authorized personnel to create, modify, publish, and manage notices with in BIT.
- **Features:**

- Text editor for notice creation (basic formatting)
- File attachment functionality (PDF, DOCX, images up to 5MB)
- Notice categorization (Academic, Event, All, Exam)
- Draft, publish, and delete functionality
- Basic scheduling for notice publication

3. Department-Based Distribution System

- **Service Offered:** Notice routing to relevant departments with in BIT according to targeting criteria.
- **Features:**
 - Department selection for notice targeting
 - Notice visibility based on user's department

4. Notification System

- **Service Offered:** Delivery of notifications to users about new or important notices.
- **Features:**
 - Email notifications for new notices (user-configurable)
 - In-application notification center
 - Priority-based display (important notices highlighted)

5. Search and Discovery System

- **Service Delivered:** Basic search capabilities to help users find notices.
- **Functionality:**
 - Keyword search in notice titles and content
 - Filtering by category and date range
 - Department-specific filtering for relevant notices
 - Most recent notices displayed by default

6. Basic Analytics

- **Service Description:** Simple data on notice engagement within BIT.
- **Features:**
 - View count tracking per notice
 - Basic usage statistics (active users, total notices)
 - Department-wise notice distribution data

7. Administration Tools

- **Service Offered:** Administration dashboard for system management at BIT.
- **Features:**
 - User management (add, remove, modify user roles)

- Department management (for notice targeting)
- System settings configuration
- Basic audit logs for security tracking

Boundary of the System

The platform is primarily intended for:

- **BIT students** seeking timely access to academic and administrative information relevant to their department and program within Bahir Dar Institute of Technology.
- **BIT faculty and staff** responsible for creating and managing official communications within their respective departments at BIT.
- **BIT administrative personnel** managing system operations and user accounts within the institute.

What the System Will Do:

- Provide digital notice creation and management for BIT
- Enable targeted notice distribution within BIT departments
- Offer basic search and filtering of notices
- Send email notifications for important announcements
- Maintain user authentication and authorization for BIT community

What the System Will Not Do:

The BIT Digital Notice Board System is **not designed to**:

- Support multiple campuses across BDU.
- Integrate with other university-wide systems during the initial deployment.
- Provide dedicated mobile applications in the initial version.
- Manage or operate emergency broadcast systems.
- Offer offline functionality for notice access.
- Include advanced analytics or reporting features.
- Manage academic courses, student records, or related administrative data.

1.9 Organization of the Project

This project is organized into three main chapters, each addressing critical aspects of the system's development and implementation for Bahir Dar Institute of Technology:

Chapter One: Introduction

This chapter introduces the project, outlining its purpose, objectives, and significance. It provides an overview of the communication inefficiencies within BIT that the platform aims to address by replacing traditional paper-based notice boards with a digital system. The chapter explains the

project's scope, limitations, and assumptions, setting the context for developing a focused solution for BIT.

Chapter Two: System Features

This chapter analyzes both the existing communication system at BIT and the proposed digital solution. It includes:

- Analysis of current physical notice board limitations at BIT
- Detailed requirements gathering and analysis
- Functional and non-functional requirements specific to BIT
- System design considerations tailored for BIT's needs
- User interface prototypes and system architecture

Chapter Three: System Design

This chapter presents the technical design of the system, including:

- System architecture and database design
- Detailed design of core modules (user management, notice management, etc.)
- Interface design and user experience considerations
- Security and authentication mechanisms
- Implementation strategy and technology stack

Chapter Two: System features

2.1 The Existing System

Currently, the process of notice dissemination at Bahir Dar Institute of Technology (BiT) is mainly carried out using traditional physical notice boards. Notices are manually prepared, printed, and posted on boards located at different buildings and departments within the institute. Students and staff access information by physically visiting these notice boards.

Although this approach has been used for many years, it lacks efficiency, accessibility, and flexibility. Notices are often placed in limited physical spaces, causing older but still relevant announcements to be removed prematurely. In many cases, important academic and administrative information does not reach the intended audience on time due to delays in printing and posting.

Furthermore, the existing system does not support targeted communication based on departments or user groups. Students who are not present on campus at the time notices are posted may completely miss important announcements. The system also lacks search functionality, making it difficult to retrieve past notices. These limitations highlight the need for a more efficient and modern digital solution for notice dissemination at Bahir Dar Institute of Technology.



Figure 2.1 Existing System

How the Existing System Functions

The existing notice dissemination system at **Bahir Dar Institute of Technology (BIT)** is predominantly **manual and paper-based**. Notices are created, approved, printed, and posted physically on notice boards located in various departments and buildings within the institute. The following steps describe how the current system operates:

1. Notice Creation and Approval

- **Manual Drafting:** Department staff prepare notices using office applications such as Microsoft Word.
- **Department-Level Approval:** Notices require approval from department heads, usually through physical signatures.
- **Paper-Based Workflow:** Approval is handled manually, which causes delays, especially when officials are unavailable.

2. Printing and Preparation

- **Printing Process:** Approved notices are printed in multiple copies depending on the number of departments.
- **Resource Consumption:** The process consumes paper, ink, and electricity.
- **Quality Issues:** Some notices become unclear due to poor printing quality.

3. Physical Posting

- **Manual Posting:** Notices are pinned or taped to physical notice boards.
- **Limited Space:** Older notices are often removed to create space, even if they are still relevant.
- **Non-Standard Placement:** Notices may be placed inconsistently across different locations.

4. Access by Students and Staff

- **Physical Presence Required:** Students must visit notice board locations to read announcements.
- **Restricted Access Hours:** Many notice boards are inside buildings accessible only during working hours.
- **Environmental Exposure:** Outdoor notice boards are affected by weather conditions.

5. Maintenance and Removal

- **Irregular Removal:** Notices are removed based on space availability rather than expiration dates.
- **Damage and Loss:** Notices may be damaged, removed, or vandalized.
- **No Backup:** Lost or damaged notices cannot be recovered.

2.1.1 Challenges of the Existing System

1. Time Inefficiency

- Notice dissemination can take **several hours or days**
- Urgent announcements are delayed
- Students waste time checking multiple notice boards

2. Resource Waste

- High paper and printing costs
- Continuous maintenance of physical boards
- Unnecessary operational expenses

3. Accessibility Limitations

- Not accessible to students off-campus
- Limited support for students with disabilities

- No access outside campus working hours

4. Information Inconsistency

- Notices may differ between departments
- Delayed posting causes confusion
- Some students never receive important information

5. Environmental Impact

- Excessive paper usage
- Contribution to environmental pollution
- Misalignment with sustainability goals

6. Security and Reliability Issues

- Notices can be removed or altered
- No tracking of who posted or viewed notices
- No audit trail

7. Lack of Analytics and Feedback

- No way to measure notice reach
- No engagement tracking
- No data for improving communication

8. Emergency Communication Limitations

- No real-time alert mechanism
- Delayed safety announcements
- Ineffective during urgent situations

2.2 Proposed System

To overcome the limitations of the existing manual system, this project proposes a **web-based Digital Notice Board System specifically for Bahir Dar Institute of Technology (BIT)**. The system is designed to modernize notice dissemination, improve accessibility, and enhance communication efficiency within the institute.

Key Features of the Proposed System

1. Digital Notice Creation and Management

- Web-based admin dashboard for creating and managing notices
- Rich text support and file attachments (PDF, images, DOCX)
- Digital approval process within the system
- Standard notice templates for consistency

2. Department-Based Notice Distribution

- Notices can be targeted to specific BIT departments
- Institute-wide notices for all users
- Controlled visibility based on user role and department

3. Web Accessibility

- Responsive web application accessible via desktop, tablet, and mobile browsers
- 24/7 access from anywhere with internet connectivity
- No requirement for physical presence on campus

4. Notification System

- Email notifications for important notices
- In-application notification center

- Highlighting of priority announcements

5. Search and Organization

- Keyword-based notice search
- Filtering by category, department, and date
- Chronological and categorized display of notices

6. Digital Archive

- Permanent storage of past notices
- Easy retrieval for reference
- Eliminates loss of information

7. Basic Analytics

- View count tracking per notice
- Basic usage statistics
- Helps administrators assess communication effectiveness

8. Security and Authentication

- Role-based access control (Student, Staff, Admin)
- Secure login using BIT institutional email
- Activity logging for accountability

9. Environmental Sustainability

- Significant reduction in paper usage
- Lower printing and maintenance costs
- Environmentally friendly communication solution

Benefits of the Proposed System

1. Efficiency Improvement

- Notice dissemination reduced from days to minutes
- Faster administrative workflows
- Improved academic coordination

2. Cost Reduction

- Reduced printing and paper costs
- Lower maintenance expenses
- Long-term operational savings

3. Enhanced Accessibility

- Anytime, anywhere access
- Inclusive for all BIT community members
- Improved access for students with disabilities

4. Improved Communication Quality

- Consistent and reliable information
- Real-time updates and corrections
- Reduced misinformation

5. Emergency Readiness

- Faster dissemination of urgent notices
- Centralized communication control
- Improved safety awareness

6. Scalability

- Can be extended to other institutes in the future
- Supports feature expansion

- Strong foundation for digital transformation

The proposed **Digital Notice Board System** represents a modern and efficient approach to institutional communication within **Bahir Dar Institute of Technology (BIT)**. By leveraging web-based technologies, the system creates a more connected, accessible, and reliable information-sharing environment for BIT students, faculty, and administrative staff.

By addressing the limitations of the existing paper-based notice boards and introducing a centralized digital platform, the proposed system significantly improves the flow of academic and administrative information within the institute. As a result, it supports timely communication, enhances administrative efficiency, and contributes to improved academic performance and institutional effectiveness at Bahir Dar Institute of Technology.

2.3 Requirement Analysis

2.3.1 Functional Requirements

FREQ-1: User Registration and Profile Management

Description: BIT students, faculty, and staff shall be able to register and manage user profiles.

Details:

- Registration using BIT institutional email (@bdu.edu.et)
- User roles: Student, Faculty/Staff, Administrator
- Profile includes: full name, ID number, department, role
- Email-based account verification
- Users can update basic profile information

Priority: High

FREQ-2: Notice Creation and Management

Description: Authorized users shall create, edit, publish, and manage notices.

Details:

- Text-based notice editor with basic formatting

- File attachments (PDF, DOCX, images up to 5MB)
- Notice categories: Academic, Event, General, Exam
- Priority levels: Normal, Important
- Publish immediately or schedule publication
- Manual or automatic notice expiration

Priority: High

FREQ-3: Department-Based Notice Distribution

Description: Notices shall be targeted to specific BIT departments.

Details:

- Department-level notice visibility
- Institute-wide notices for all BIT users
- Role-based restrictions on notice publishing

Priority: High

FREQ-4: Notice Viewing and Access

Description: Users shall view notices easily through the system.

Details:

- Chronological display of notices
- Detailed notice view with attachments
- Highlight important notices
- 24/7 web access

Priority: High

FREQ-5: Search and Filtering

Description: Users shall search and filter notices.

Details:

- Keyword search (title and content)
- Filter by category, department, and date
- Sort by newest and priority

Priority: High

FREQ-6: Notification System

Description: Users shall receive notifications for new notices.

Details:

- Email notifications for important notices
- In-application notification center
- User preference for notifications

Priority: Medium

FREQ-7: User Role and Permission Management

Description: System shall enforce role-based access control.

Details:

- Administrator: full system control
- Faculty/Staff: create notices (subject to approval)
- Student: view notices only
- Role assignment managed by administrator

Priority: High

FREQ-8: Notice Approval Workflow

Description: Important notices shall require approval before publishing.

Details:

- Status states: Draft, Pending Approval, Published
- Admin approval for official notices
- Notification of approval decisions

Priority: Medium

FREQ-9: Digital Notice Archive

Description: System shall store past notices securely.

Details:

- Archived notices remain searchable
- Prevents information loss
- Supports institutional record keeping

Priority: Medium

FREQ-10: Basic Analytics

Description: System shall track basic usage statistics.

Details:

- View count per notice
- Total notices published
- User activity overview

Priority: Low

FREQ-11: Responsive Web Interface

Description: System shall work on different devices.

Details:

- Responsive design for desktop, tablet, and mobile browsers
- Cross-browser compatibility

Priority: High

FREQ-12: Security and Authentication

Description: System shall protect user data and notices.

Details:

- Secure login with email and password
- Password encryption
- Session management
- Access control enforcement

Priority: High

FREQ-1: User Account Registration and Profile Management

Description:

Students and staff of Bahir Dar Institute of Technology shall be able to create accounts and manage their profiles.

Details:

- Registration using official institutional email addresses ending with @bdu.edu.et
- User profile information shall include:
 - Full name
 - ID number
 - Department
 - Program and year of study
- Users shall manage notification preferences and privacy settings
- Account verification through institutional email confirmation

Priority: High

FREQ-2: Notice Creation and Management

Description:

Authorized BIT staff shall create, edit, schedule, and manage digital notices.

Details:

- Rich text editor with formatting options
- Attachment support (PDF, DOCX, images up to 10 MB)
- Department-based targeting within BIT
- Notice categories:
 - Academic
 - Event
 - Emergency
 - Announcement
 - General
 - Exam
- **Priority levels:** Low, Normal, High, Urgent
- Scheduling and automatic expiration of notices

Priority: High

FREQ-3: Advanced Search and Filtering

Description:

The system shall provide efficient search and filtering for BIT notices.

Details:

- Full-text search across notice titles and content
- Filters by department, category, date, and priority

- Keyword highlighting in results

Priority: High

FREQ-4: Real-Time Notification System

Description:

The system shall notify BIT users of new and important notices.

Details:

- Email notifications
- In-application notifications
- SMS alerts for urgent and emergency notices
- Custom notification preferences

Priority: High

FREQ-5: Multi-Language Support

Description:

The system shall support English and Amharic languages.

Details:

- Language toggle for system interface
- Bilingual notice display

Priority: Medium

FREQ-6: Analytics and Reporting

Description:

BIT administrators shall monitor notice performance.

Details:

- View counts and engagement statistics

- Department-level usage reports
- Export reports in PDF and Excel formats

Priority: Medium

FREQ-7: Offline Access Capability

Description: The system shall allow limited offline access on mobile devices.

Details:

- Cache recently viewed notices
- Sync data when internet access is restored

Priority: Medium

FREQ-8: Accessibility Features

Description: The system shall be accessible to all BIT users.

Details:

- Screen reader compatibility
- Adjustable text size and contrast
- Keyboard navigation support

Priority: Medium

FREQ-9: Comment and Feedback System

Description:

Users shall interact with notices through feedback.

Details:

- Commenting and reactions
- Reporting inappropriate content

Priority: Low

FREQ-10: Emergency Broadcast System

Description:

The system shall broadcast emergency alerts within BIT.

Details:

- Emergency notices override normal settings
- Instant alerts to all users

Priority: High

FREQ-11: Document Management

Description:

The system shall securely manage notice attachments.

Details:

- File upload restrictions
- Document preview and version control

Priority: Medium

FREQ-12: User Roles and Permission Management

Description:

The system shall enforce role-based access control.

Details:

- System Administrator
- Department Administrator
- Staff

- Student

Priority: High

FREQ-13: Notice Approval Workflow

Description:

Important notices shall require approval before publication.

Details:

- Draft → Pending → Approved → Published workflow
- Approval notifications and tracking

Priority: Medium

FREQ-14: Calendar Integration

Description:

The system shall integrate notices with calendars.

Details:

- Event reminders
- Calendar export support

Priority: Low

FREQ-15: Social Sharing

Description:

The system shall allow controlled sharing of notices.

Details:

- Shareable links
- QR code generation

Priority: Low

FREQ-16: Audit Trail and Activity Logging

Description:

The system shall log all major activities.

Details:

- User login and notice activity logs
- Exportable audit reports

Priority: Medium

FREQ-17: Bulk Operations

Description:

The system shall support bulk notice management.

Details:

- Bulk creation and archiving

Priority: Low

FREQ-18: Responsive Design

Description:

The system shall function on all devices.

Details:

- Mobile-first design
- Cross-browser compatibility

Priority: High

FREQ-19: Data Export and Backup

Description:

The system shall ensure data safety and recovery.

Details:

- Data export options
- Automated and manual backups

Priority: Medium

2.3.2 System Use case

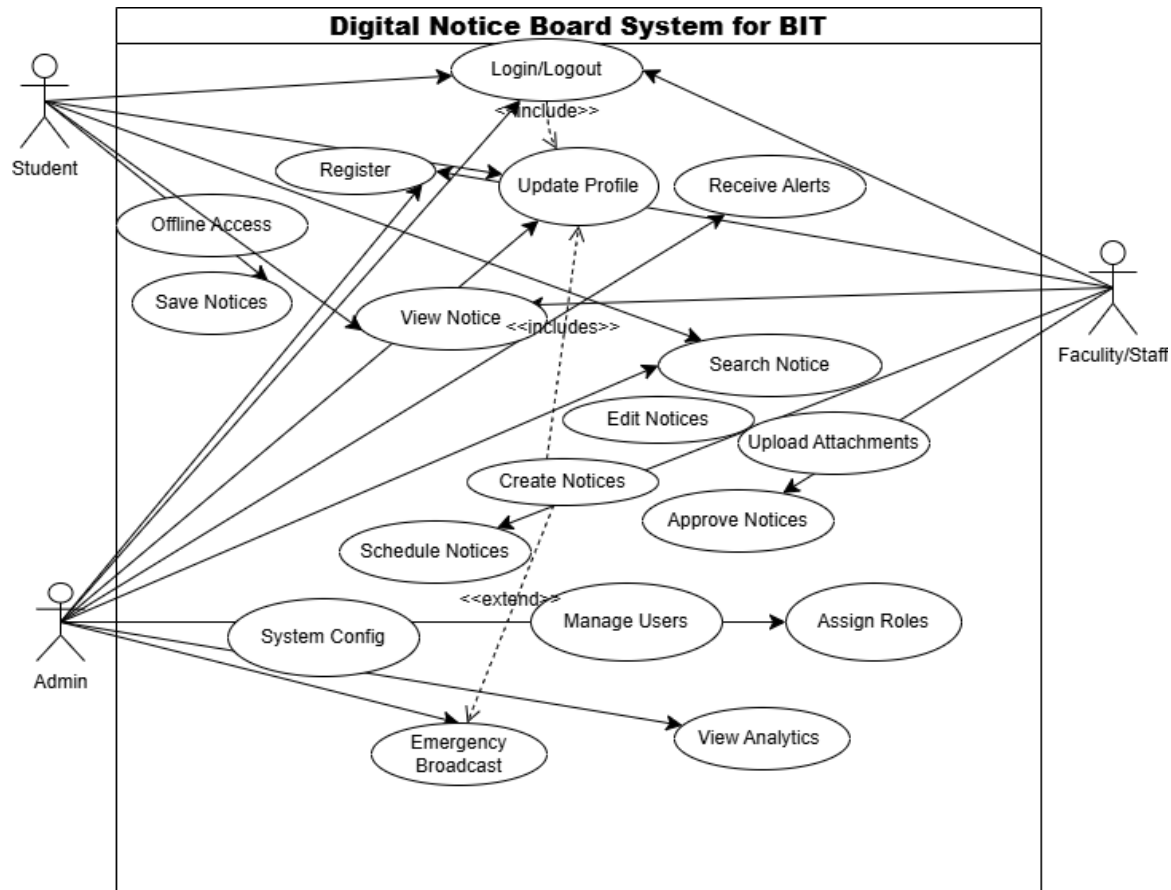


Figure 2.2 use case diagram

2.3.2.1 Use Case documentation

Table 2.1: Description of register use case

Use Case ID	001
Use Case Name	User Registration
Participating Actor	• Student • Staff/Faculty

Use Case Description	Allows new users to create an account in the system with appropriate role assignment.
Pre-condition	Users not registered
Flow of Events	1. User navigates to registration page. 2. User selects role (Student, Staff, etc.). 3. User fills in required details: Name, Email, Password, Department, ID Number. 4. User clicks "Register". 5. System sends verification email. 6. User clicks verification link in email. 7. System activates account and assigns default permissions.
Alternate Flow	3.1 If email already registered, system shows error. 6.1 If verification link expires, user can request new verification email.
Post Condition	User account is created and ready for login.

Table 2.2: Description of Login/Logout Use Case

Use Case ID	002
Use Case Name	Login
Participating Actor	● Student ● Staff/Faculty ● Admin
Use Case Description	Allows users to securely access and exit the system.
Pre-condition	User must have a registered.
Flow of Events	1. User enters credentials. 2. System verifies. 3. Access granted.

Alternate Flow	4.1 If credentials invalid, system shows "Invalid email or password". 4.2 If account is locked, system shows appropriate message.
Post Condition	User is logged in or logged out successfully.

Table 2.3: Description of View Notices Use Case

Use Case ID	003
Use Case Name	View Notices
Participating Actor	● Student ● Staff/Faculty
Use Case Description	Allows users to view all published notices on the notice board.
Pre-condition	User must be logged in.
Flow of Events	1. User navigates to the notice board page. 2. System displays a list of published notices. 3. User clicks on a notice to view its details. 4. System shows the notice content, date, attachments, and comments.
Alternate Flow	2.1 If no notices are available, the system displays "No notices found."
Post Condition	User successfully views the notice.

Table 2.4: Description of Filter/Search Notices Use Case

Use Case ID	004
Use Case Name	Filter/Search Notices
Participating Actor	● Student ● Staff/Faculty

Use Case Description	Allows users to search or filter notices by category, date, or keywords.
Pre-condition	User must be logged in.
Flow of Events	1. User goes to notice board. 2. User uses search bar or filter options. 3. System displays filtered results. 4. User selects notice from results.
Alternate Flow	2.1 If no results found, system shows "No matching notices."
Post Condition	User views filtered notices.

Table 2.5: Description of Create/Edit Notice Use Case

Use Case ID	005
Use Case Name	Create/Edit Notice
Participating Actor	● Staff/Faculty
Use Case Description	Allows authorized users to create or modify notice.
Pre-condition	Faculty must be logged in and have permission to create notices.
Flow of Events	1. Faculty clicks "Create Notice". 2. System opens a notice creation form. 3. Faculty fills in title, content, and attaches files if needed. 4. Faculty submits the notice. 5. System saves the notice as draft or sends it for approval.
Alternate Flow	4.1 If required fields are missing, system shows validation error.
Post Condition	Notice is created and saved in the system.

Table 2.6: Description of Upload Attachments Use Case

Use Case ID	006
Use Case Name	Upload Attachments
Participating Actor	● Staff/Faculty
Use Case Description	Allows authorized users to add, remove, or update attachments for a notice.
Pre-condition	Notice creation/editing is active.
Flow of Events	1. Faculty/Staff opens a notice. 2. Faculty/Staff clicks "Upload Attachments". 3. System shows attachment interface. 4. Faculty/Staff uploads new files or deletes existing ones. 5. Faculty/STAFF saves changes.
Alternate Flow	4.1 If file size exceeds limit, system shows error.
Post Condition	Attachments are updated for the notice.

Table 2.7: Description of Approve Notices Use Case

Use Case ID	007
Use Case Name	Approve Notices
Participating Actor	● Administrator
Use Case Description	Allows authorized users to review and approve pending notices.
Pre-condition	Notice is submitted
Flow of Events	1. Faculty/Staff navigates to "Approve Notices" section. 2. System/Staff shows list of pending notices. 3. Faculty/Staff reviews a notice. 4. Faculty/Staff approves or rejects the notice. 5. System updates notice status and notifies creator.
Alternate Flow	4.1 If rejected, faculty/staff can add a reason for rejection.
Post Condition	Notice becomes visible to users.

Table 2.8: Description of Emergency Broadcast Use Case

Use Case ID	008
Use Case Name	Emergency Broadcast
Participating Actor	● Administrator
Use Case Description	Allows admin to send urgent notices to all users immediately.
Pre-condition	Admin must be logged in.
Flow of Events	1. Admin clicks "Emergency Broadcast". 2. System opens emergency notice form. 3. Admin enters urgent message. 4. Admin sends broadcast. 5. System immediately publishes notice to all users.
Alternate Flow	3.1 If message is empty, system prevents sending.
Post Condition	Emergency notice is published to all users instantly.

Table 2.9: Description of Manage Users/Roles Use Case

Use Case ID	009
Primary Actor	Administrator
Description	Manages user accounts and assigns roles.
Precondition	Admin access required.
Postcondition	User roles are updated.
Main Flow	1. Admin selects user. 2. Assigns role

Table 2.10: Description of View User Profile/Edit Use Case

Use Case ID	018
Use Case Name	View User Profile
Participating Actor	● All Users

Use Case Description	Allows users to view and update their own profile information.
Pre-condition	User must be logged in.
Flow of Events	1. User clicks on their profile icon/name. 2. System displays profile page with user details: Name, Email, Role, Department, Registration Date. 3. User can view and updates details
Post Condition	User views/updated their profile information.

Table 2.11: Description of Delete Notice Use Case

Use Case ID	020
Use Case Name	Delete Notice
Participating Actor	● Faculty/Staff ● Administrator
Use Case Description	Allows authorized users to permanently delete notices from the system.
Pre-condition	User must have delete permissions; notice must not be archived.
Flow of Events	1. User selects a notice to delete. 2. User clicks "Delete Notice". 3. System asks for confirmation with warning about permanent deletion. 4. User confirms deletion. 5. System removes notice from database and all references.
Alternate Flow	4.1 If user cancels, system returns to notice list without changes.
Post Condition	Notice is permanently deleted from the system.

Table 2.12: Description of Notice Archiving /Restore Use Case

Use Case ID	014
Use Case Name	Notice Archiving/Restore
Participating Actor	● Administrator
Use Case Description	Allows administrators to archive old notices to

	maintain system performance and organization and restores them when required
Pre-condition	User is authorized
Flow of Events	1. Select notice. 2. Archive or restore. 3. System confirms action.
Alternate Flow	4.1 If notices are still active/pinned, system prevents archiving with warning.
Post Condition	Selected notices are archived/restored and removed from active display.

2.3.3 Business Rule Documentation

[BR-1]

- **Identifier:** BR-1
- **Name:** Institutional Email Requirement
- **Description:** All users must register with a valid institutional email address (@bdu.edu.et) to access the system.

[BR-2]

- **Identifier:** BR-2
- **Name:** Unique User Account
- **Description:** Each user can have only one active account per institutional email address. Duplicate accounts will be deactivated.

[BR-3]

- **Identifier:** BR-3
- **Name:** Role-Based Registration

- **Description:** During registration, users must select their correct role (Student, Staff/Faculty, or Administrator) which determines their system access and permissions.

[BR-4]

- **Identifier:** BR-4
- **Name:** Email Verification Requirement
- **Description:** All accounts must be verified through email confirmation before gaining access to system features.

[BR-5]

- **Identifier:** BR-5
- **Name:** Password Security Policy
- **Description:** Passwords must be at least 8 characters long, containing uppercase letters, lowercase letters, numbers, and special characters.

[BR-6]

- **Identifier:** BR-6
- **Name:** Account Lockout Policy
- **Description:** After 5 consecutive failed login attempts, the account will be locked for 30 minutes. Account recovery requires email verification.

[BR-7]

- **Identifier:** BR-7
- **Name:** Notice Creation Privilege
- **Description:** Only Staff/Faculty and Administrators can create notices. Students can only view notices.

[BR-8]

- **Identifier:** BR-8

- **Name:** Mandatory Notice Approval
- **Description:** All notices created by Staff/Faculty must be reviewed and approved by an Administrator before publication.

[BR-9]

- **Identifier:** BR-9
- **Name:** Department-Specific Notices
- **Description:** Notices can be targeted to specific BIT departments. Users only view notices relevant to their department unless marked as institute-wide.

[BR-10]

- **Identifier:** BR-10
- **Name:** Edit Restriction Policy
- **Description:** Once a notice is published, only the original creator or an Administrator can edit it. Edited notices require re-approval.

[BR-11]

- **Identifier:** BR-11
- **Name:** Attachment Size Limit
- **Description:** Notice attachments must not exceed **5MB per file**, with a maximum of **3 attachments per notice**.

[BR-12]

- **Identifier:** BR-12
- **Name:** Notice Retention Policy
- **Description:** Notices are automatically archived after a defined period or manually archived by an Administrator.

[BR-13]

- **Identifier:** BR-13
- **Name:** Role-Based Access Control
- **Description:** System features are accessible based on assigned user roles:
 - **Student:** View notices only
 - **Faculty/Staff:** Create and manage notices
 - **Administrator:** Full system control

[BR-14]

- **Identifier:** BR-14
- **Name:** Notification Delivery Rules
- **Description:** Important notices trigger email notifications to relevant users based on department and role.

[BR-15]

- **Identifier:** BR-15
- **Name:** Audit Trail Maintenance
- **Description:** All notice creation, approval, editing, and deletion actions are logged with user and timestamp information.

[BR-16]

- **Identifier:** BR-16
- **Name:** Emergency Broadcast Authorization
- **Description:** Only Administrators can send emergency broadcasts, which bypass normal approval workflows.

[BR-17]

- **Identifier:** BR-17
- **Name:** Notice Priority Levels
- **Description:** Notices must be assigned a priority level (Low, Medium, High, Emergency) which determines display order and notification methods.

[BR-18]

- **Identifier:** BR-18
- **Name:** Session Timeout Policy
- **Description:** User sessions automatically expire after 30 minutes of inactivity, requiring re-authentication.

[BR-19]

- **Identifier:** BR-19
- **Name:** Data Privacy Compliance
- **Description:** All user data must be handled in accordance with institutional data privacy policies and relevant data protection regulations.

[BR-20]

- **Identifier:** BR-20
- **Name:** Notice Deletion Authority
- **Description:** Administrators can permanently delete notices.

[BR-21]

- **Identifier:** BR-21
- **Name:** Accessibility Compliance
- **Description:** All notices and system interfaces must comply with web accessibility standards (WCAG 2.1 AA) to ensure usability for users with disabilities.

[BR-22]

- **Identifier:** BR-22
- **Name:** Multi-language Support Requirement
- **Description:** All system-critical notices must be available in both English and Amharic languages.

[BR-23]

- **Identifier:** BR-23
- **Name:** System Maintenance Window
- **Description:** Scheduled system maintenance will only occur during non-peak hours (10:00 PM - 5:00 AM local time) with 48 hours advance notice to all users.

2.3.4 User Interface prototype

Student User Interface Prototype - BIT Digital Notice Board

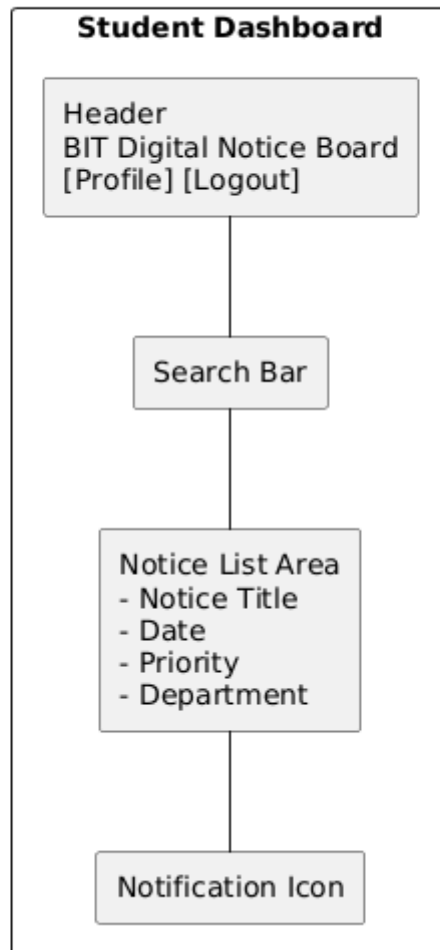


Figure 2.3: Student UI prototype

Staff/Faculty User Interface Prototype - BIT Digital Notice Board

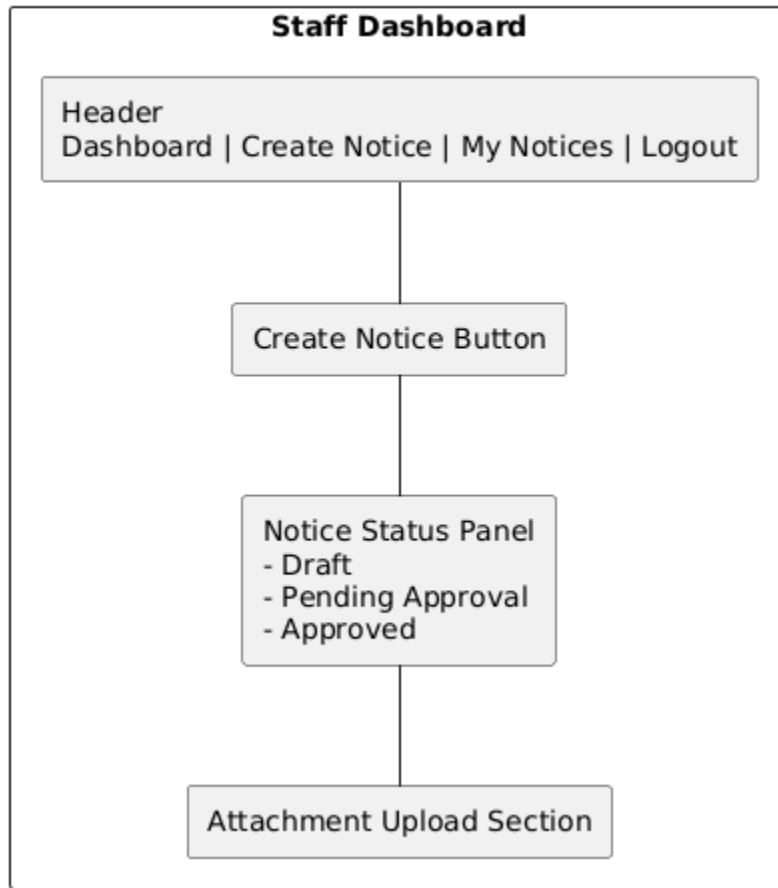


Figure 2.4: Faculty/Staff UI prototype

Administrator User Interface Prototype - BIT Digital Notice Board

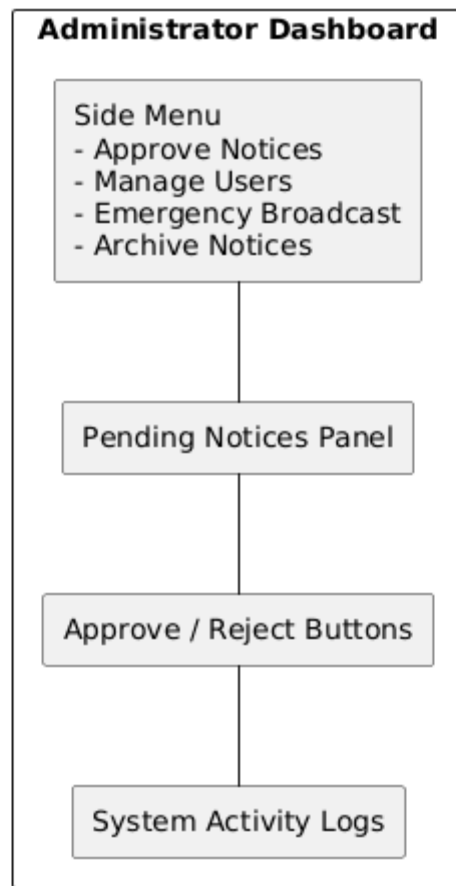


Figure 2.5: Admin UI prototype

2.3.5 State chart diagram

Login State Chart - Digital Notice Board (BIT)

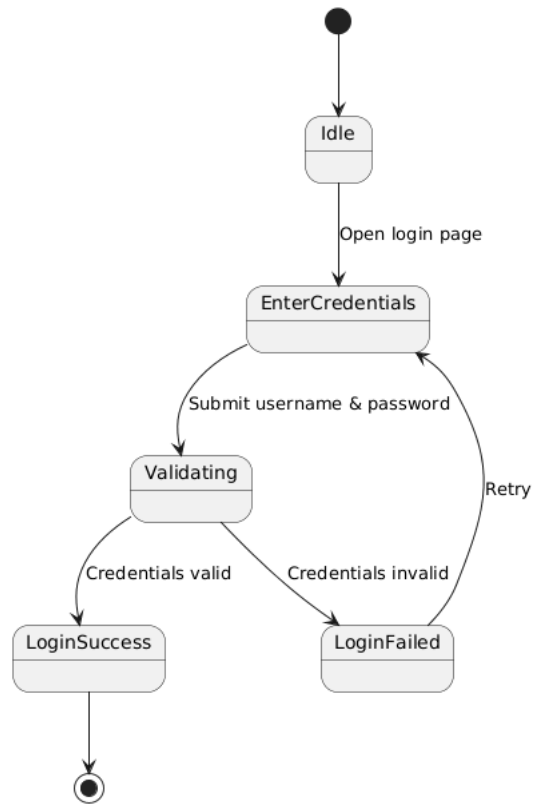


Figure 2.6: Login State Chart diagram

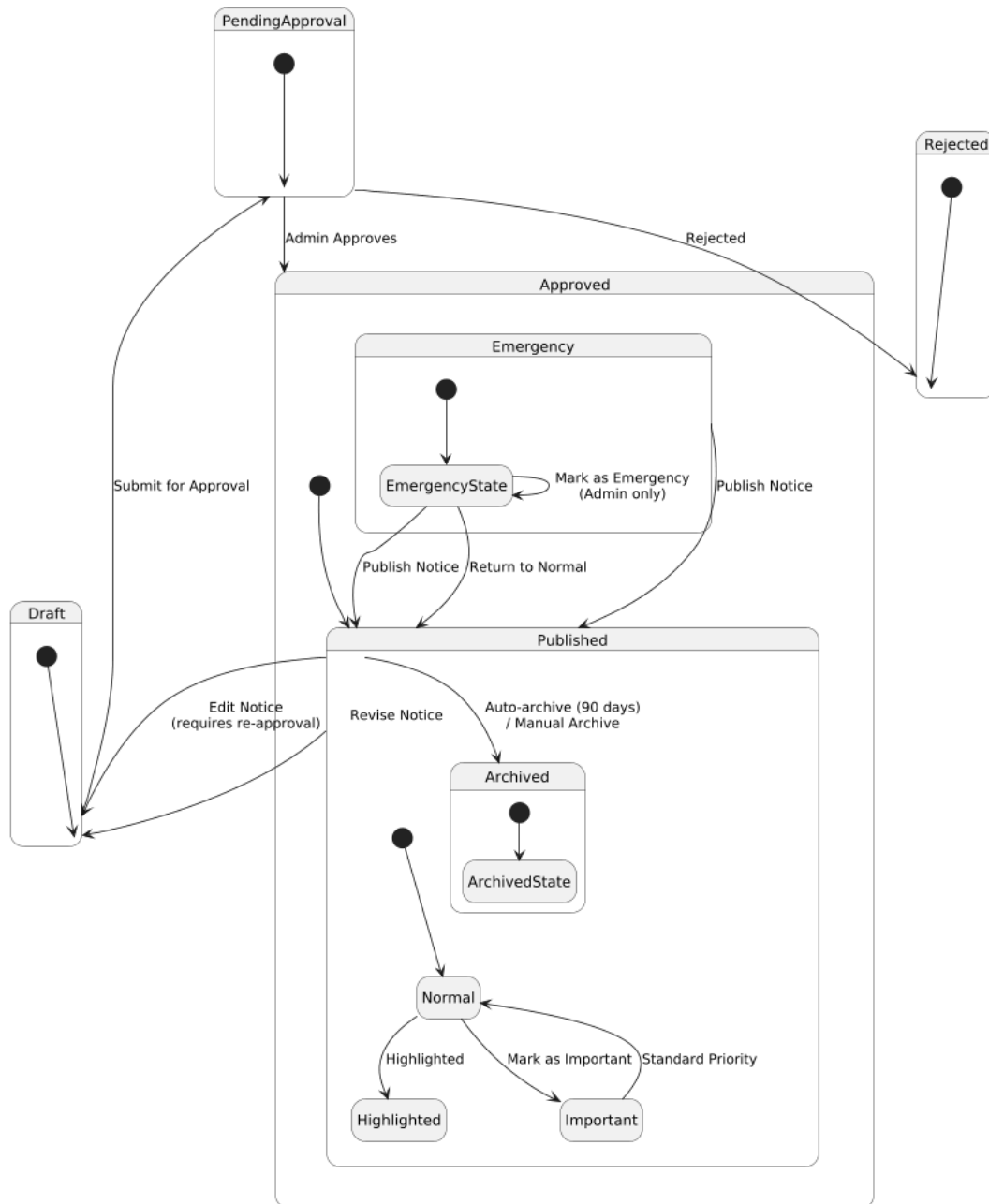


Figure 2.7: Notice Creation & Approval State Chart diagram

2.3.6 Activity Diagram

Registration Activity Diagram - Digital Notice Board (BIT)

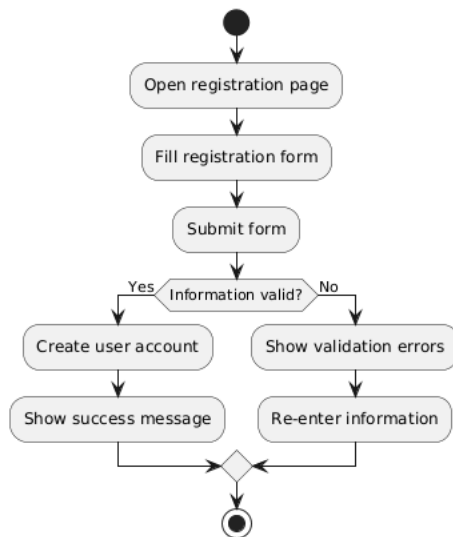


Figure 2.8: Registration Activity Diagram

Login Activity Diagram - Digital Notice Board (BIT)

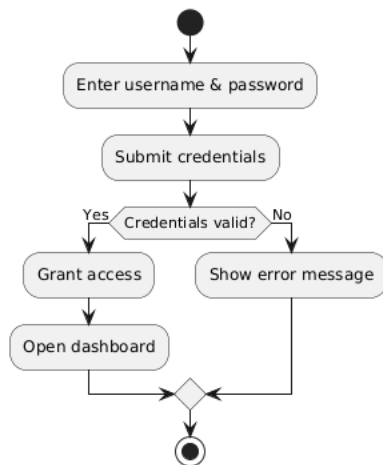


Figure 2.9: Login Activity Diagram

Notice Creation Activity Diagram - Digital Notice Board (BIT)

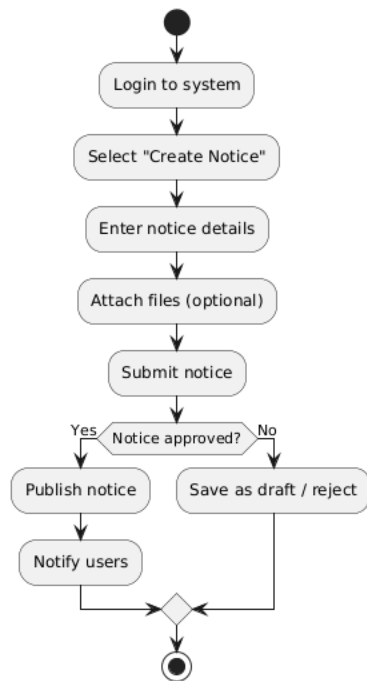


Figure 2.10: Create Notice Activity Diagram

2.3.7 Sequence diagram

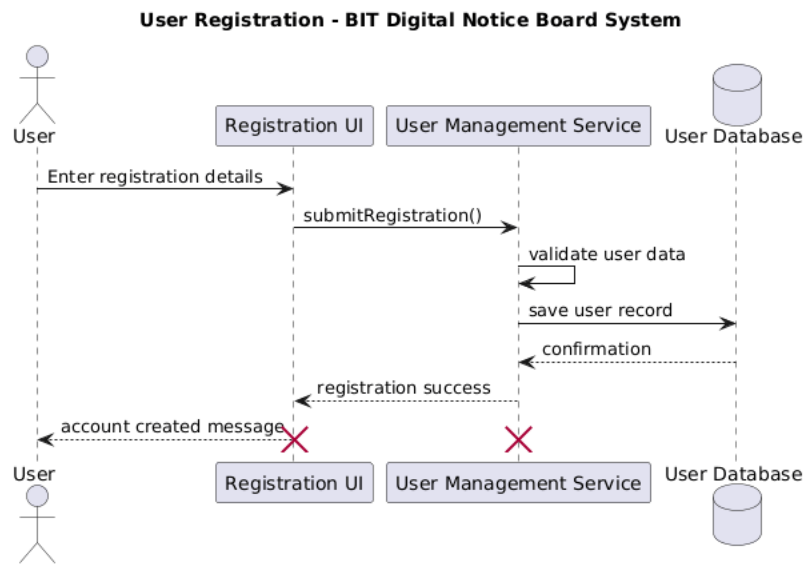


Figure 2.11: Registration Sequence Diagram

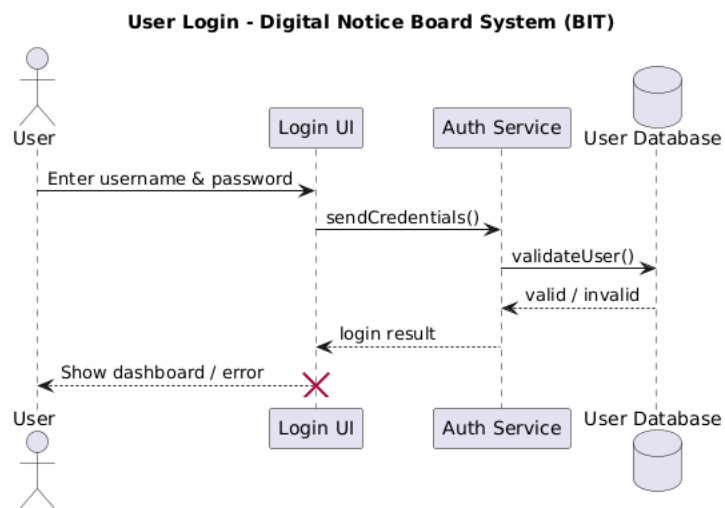


Figure 2.12: Login Sequence Diagram

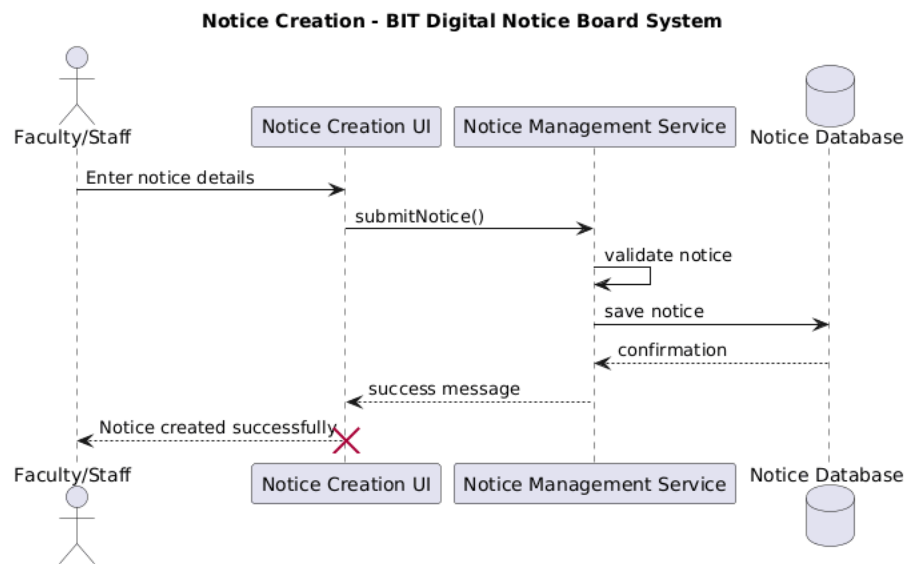


Figure 2.13: Notice Creation Sequence Diagram

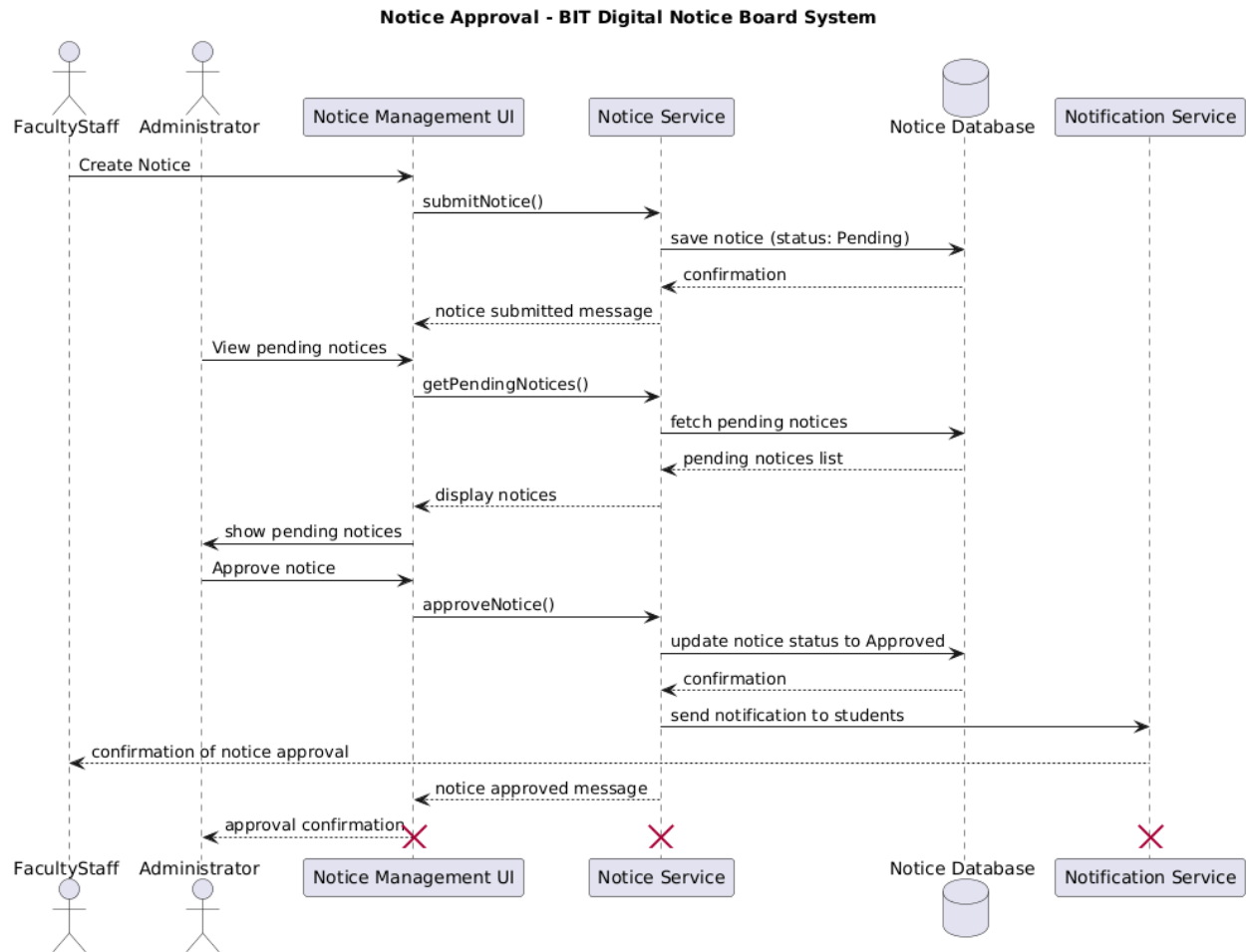


Figure 2.14: Notice Approval Sequence Diagram

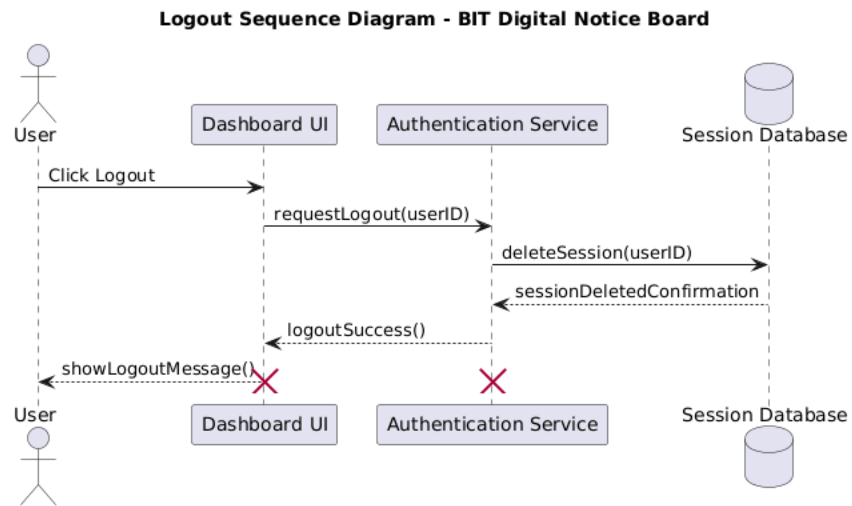


Figure 2.15: Logout Sequence Diagram

2.3.8 Analysis Class Model

Classes

User:

- Attributes: userID, fullName, email (@bdu.edu.et), password, phone, department, role, registrationDate, lastLoginDate, status
- Methods: register(), login(), logout(), updateProfile(), viewProfile(), changePassword(), resetPassword(), setNotificationPreferences()

Student (Inherits User):

- Attributes: studentID, program, yearOfStudy
- Methods: viewNotices(), searchNotices(), filterNotices(), likeNotice(), dislikeNotice(), commentOnNotice(), reportNotice(), saveNotice(), getSavedNotices()

Faculty/Staff (Inherits User):

- Attributes: staffID, designation, officeNumber, canCreateNotices, maxAttachmentSize, allowedNoticeCategories
- Methods: createNotice(), editNotice(), deleteNotice(), submitForApproval(), attachFiles(), scheduleNotice(), manageNoticeTemplates(), trackNoticePerformance()

Administrator (Inherits User):

- Attributes: adminID, accessLevel, canApproveNotices, canManageUsers, canArchiveNotices
- Methods: approveNotice(), rejectNotice(), manageUsers(), assignRoles(), archiveNotice(), viewAnalytics(), generateReports()

Notice:

- Attributes: noticeID, title, content, category (Academic, Event, Emergency, Announcement, General, Exam), priority (Low, Normal, High, Urgent), status (Draft, Pending Approval, Published, Archived), createdBy, createdAt, approvedBy, approvedAt,

publishedAt, expiresAt, attachments[], targetDepartments[], viewCount, likeCount, commentCount

- Methods: publish(), archive(), updateContent(), schedule(), extendExpiry(), trackViews(), incrementLikeCount(), incrementCommentCount(), attachFile(), validateNotice(), notifyTargetUsers()

Attachment:

- Attributes: attachmentID, noticeID, fileName, fileType (PDF, DOCX, Image), fileSize, uploadDate, version, downloadCount
- Methods: uploadFile(), downloadFile(), previewFile(), validateFileType(), compressFile(), trackDownloadCount()

Comment:

- Attributes: commentID, noticeID, userID, content, timestamp, likes, dislikes, parentCommentID, status, editHistory, lastEditedAt
- Methods: addComment(), editComment(), deleteComment(), likeComment(), dislikeComment(), replyToComment(), reportComment()

Notification:

- Attributes: notificationID, userID, noticeID, type (email, SMS, in-app), message, sentAt, readStatus, deliveryMethod
- Methods: sendNotification(), markAsRead(), deleteNotification(), trackDelivery()

AuditLog:

- Attributes: logID, userID, action, timestamp, details, moduleAffected, severityLevel
- Methods: logAction(), viewLogs(), exportLogs(), filterLogs()

Department:

- Attributes: departmentID, name, totalUsers, totalNotices
- Methods: getDepartmentNotices(), validateDepartment(), calculateStatistics()

ApprovalWorkflow:

- **Attributes:** workflowID, noticeID, approverID, status (Draft, Pending Approval, Approved, Published), comments, createdAt, updatedAt, deadline
- **Methods:** initiateApproval(), approveStep(), rejectStep(), trackProgress(), notifyApprover()

Report:

- **Attributes:** reportID, reporterID, reportedItemID, reportedItemType, reason, description, timestamp, status, resolvedBy, resolution, resolvedAt, severity
- **Methods:** submitReport(), updateReport(), resolveReport(), escalateReport(), viewReport(), generateReportSummary(), exportReports(), notifyModerator(), trackReportStatus()

Relationships

- **Student:**
 - Can view multiple notices, download attachments, comment, like/dislike, save notices
 - Belongs to one Department
 - Receives notifications
- **Faculty/Staff:**
 - Can create, edit, submit notices for approval
 - Manages attachments
 - Belongs to one Department
 - Can receive approval notifications
- **Administrator:**
 - Approves/rejects notices

- Manages all users and their roles
 - Archives notices
 - Generates analytics reports
- **Notice:**
 - Belongs to one Department (or multiple for institute-wide)
 - Has multiple Comments, Attachments, Notifications
 - Follows an ApprovalWorkflow for important notices
- **Comment:**
 - Belongs to one Notice and one User
 - Can have replies (self-referencing)
- **Attachment:**
 - Belongs to one Notice
 - Can be downloaded by multiple Users
- **Notification:**
 - Belongs to one User
 - References one Notice

2.3.9 Logic Model

LOGIC MODEL: USER REGISTRATION

BEGIN User_Registration

INPUT: Name, Email, Phone_Number, Password, User_Role, Department, ID_Number

LOAD Registration_Page

USER_ACTION: Enter registration details

USER_ACTION: Click Register button

IF (Form_Is_Filled) THEN

IF (Email_Ends_With("@bit.edu.et")) THEN

IF (Valid_Input_Values) THEN

IF (NOT Email_ID_Exists_In_Database) THEN

Hash_Password()

Generate_Verification_Token()

Save_To_Database(User_Role)

Send_Verification_Email()

RETURN "Registration successful. Check email for verification link"

ELSE

RETURN "Email or ID Number already exists in system"

ENDIF

ELSE

RETURN "Invalid input values. Please check your information"

ENDIF

```

ELSE

    RETURN "Only institutional email addresses (@bit.edu.et) are allowed"

ENDIF

ELSE

    RETURN "Empty submission, please fill all required fields"

ENDIF

END User_Registration

LOGIC MODEL: USER LOGIN

BEGIN User_Login

    INPUT: Email/Username, Password

    LOAD Login_Page

    USER_ACTION: Enter credentials

    USER_ACTION: Click Login button

    IF (Form_Is_Filled) THEN

        IF (Valid_Input_Values) THEN

            IF (Credentials_Match_Database) THEN

                IF (Account_Is_Verified) THEN

                    IF (Account_Is_Not_Locked) THEN

                        Generate_Session_Token()

                        Update_Last_Login_Timestamp()

                        Reset_Failed_Login_Attempts()

                        Redirect_To_Dashboard(User_Role)

                        RETURN "Login successful"

```

```

ELSE

    RETURN "Account is locked. Please reset password or contact admin"

ENDIF

ELSE

    RETURN "Account not verified. Please check your email for verification link"

ENDIF

ELSE

    Increment_Failed_Login_Attempts()

    IF (Failed_Attempts >= 5) THEN

        Lock_Account_For_30_Minutes()

        Send_Security_Alert_Email()

        RETURN "Account locked due to multiple failed attempts"

    ELSE

        RETURN "Invalid credentials. Attempts remaining: [5 - Failed_Attempts]"

    ENDIF

ENDIF

ELSE

    RETURN "Invalid input values"

ENDIF

ELSE

    RETURN "Empty submission, please enter email and password"

ENDIF

END User_Login

```

LOGIC MODEL: CREATE NOTICE

BEGIN Create_Notice

INPUT: Title, Content, Category, Priority, Target_Departments, Attachments, Schedule_Date,
Expiry_Date

LOAD Create_Notice_Page

USER_ACTION: Enter notice details

USER_ACTION: Click Submit for Approval button

IF (Form_Is_Filled) THEN

IF (Title_And_Content_Not_Empty) THEN

IF (User_Has_Permission_To_Create) THEN

IF (Attachments_Are_Valid) THEN

Save_Notice_As_Draft()

IF (Emergency_Notice_Selected) THEN

Set_Notice_Status("Emergency Pending")

Require_Dual_Authorization()

Notify_Emergency_Approvers()

RETURN "Emergency notice submitted for dual authorization"

ELSE

Determine_Appropriate_Approver(Department, Priority_Level)

Send_Approval_Request()

Set_Notice_Status("Pending Approval")

RETURN "Notice submitted for approval. Approver: [Approver_Name]"

ENDIF

```

ELSE

    RETURN "Invalid attachments. Check file size and type restrictions"

ENDIF

ELSE

    RETURN "You don't have permission to create notices"

ENDIF

ELSE

    RETURN "Notice title and content are required"

ENDIF

ELSE

    RETURN "Please fill all required fields"

ENDIF

END Create_Notice

LOGIC MODEL: APPROVE NOTICE

BEGIN Approve_Notice

    INPUT: Notice_ID, Approval_Decision, Comments

    LOAD Approval_Dashboard

    USER_ACTION: Select notice to review

    USER_ACTION: View notice details

    IF (User_Has_Approval_Permissions) THEN

        IF (Decision_Needed) THEN

            CASE Approval_Decision OF

                "Approve":

```

```

IF (Approver_Has_Required_Authority) THEN

    Update_Notice_Status("Approved")

    Set_Approval_Timestamp()

    IF (Notice_Has_Scheduled_Publish_Date) THEN

        Schedule_For_Automatic_Publishing()

        RETURN "Notice approved and scheduled for [Publish_Date]"

    ELSE

        Publish_Notice_Immediately()

        Send_Notifications_To_Target_Departments()

        Update_Notice_Board()

        RETURN "Notice approved and published immediately"

    ENDIF

ELSE

    RETURN "You don't have authority to approve this notice level"

ENDIF

"Reject":

    Enter_Rejection_Reason()

    Update_Notice_Status("Rejected")

    Send_Rejection_Notification_To_Creator()

    RETURN "Notice rejected. Creator notified"

"Request_Changes":

    Enter_Change_Requests()

    Update_Notice_Status("Changes Requested")

```



```

        Send_Change_Request_To_Creator()

        RETURN "Change request sent to creator"

    ENDCASE

ELSE

    RETURN "No decision made"

ENDIF

ELSE

    RETURN "You don't have permission to approve this notice"

ENDIF

END Approve_Notice

LOGIC MODEL: SEARCH AND FILTER NOTICES

BEGIN Search_And_Filter_Notices

    INPUT: Search_Keywords, Department_Filter, Category_Filter, Date_Range, Priority_Level

    LOAD Notice_Board_Page

    USER_ACTION: Enter search terms or select filters

    USER_ACTION: Click Search button

    IF (Search_Or_Filters_Applied) THEN

        IF (Valid_Search_Parameters) THEN

            Build_Query_With_Applied_Filters()

            IF (User_Is_Student) THEN

                Filter_By_Users_Department()

            ELSEIF (User_Is_Staff) THEN

                Include_Department_Notices()

```

```

ELSEIF (User_Is_Admin) THEN

    Include_All_Notices()

ENDIF

Sort_Results_By(

    Priority: Emergency > High > Normal > Low,

    Date: Newest_First

)

Paginate_Results(20_per_page)

Display_Results_With_Highlighting()

Log_Search_Activity()

RETURN "Search completed. Found [X] results"

ELSE

    RETURN "Invalid search parameters. Please refine your search"

ENDIF

ELSE

    Display_Default_Notices(

        Based_On: User_Role,

        User_Department,

        Priority_And_Recency

    )

    RETURN "Showing recent notices for your department"

ENDIF

END Search_And_Filter_Notices

```

LOGIC MODEL: GENERATE REPORT

BEGIN Generate_Report

INPUT: Report_Type, Date_Range, Department_Filter, Output_Format

LOAD Reports_Page

USER_ACTION: Select Report_Type from:

- Usage_Statistics
- Performance_Metrics
- User_Activity
- Notice_Analytics

USER_ACTION: Set Date_Range

USER_ACTION: Select Department (if applicable)

USER_ACTION: Select Output_Format (PDF, Excel, CSV)

USER_ACTION: Click Generate Report button

IF (User_Has_Report_Generation_Permissions) THEN

IF (Valid_Parameters_Selected) THEN

Query_Database_For_Report_Data()

Process_And_Format_Data()

Generate_Report_In_Selected_Format()

Log_Report_Generation()

Provide_Download_Link()

RETURN "Report generated successfully. Click to download"

ELSE

RETURN "Invalid report parameters"

```
ENDIF  
  
ELSE  
  
    RETURN "You don't have permission to generate reports"  
  
ENDIF  
  
END Generate_Report
```

2.4 Non-Functional Requirements

2.4.1 User Interface and Human Factors

The Digital Notice Board System will have an intuitive and accessible user-friendly interface, which can easily be adapted for both the less or the highly computer-literate user. This interface should be characterized by simplicity and clarity so that it is easy for students, staff, and administrators to get their way around this system with minimal training.

Key Interface Requirements:

Predictable Navigation: Standard navigation patterns throughout all modules and user roles

Responsive Design: Seamless experience across desktop devices ranging from 1920x1080 to 1366x768, tablet 1024x768, and mobile devices ranging from 360x640 to 414x896.

Intuitive design: Logically group related functions, use a clear visual hierarchy.

Accessibility Compliance: WCAG 2.1 Level AA compliance for users with disabilities

Localized Interface: English and Amharic It would also support RTL for the Amharic text.

Progressive Disclosure: Sophisticated features unveiled piece by piece as users progress.

Visual Feedback: Instant feedback for the user's action such as hover, click, submit form.

Consistent Iconography: Standardized sets of icons are used, and there are tooltips for clarity.

Readable Typography: Body text with the minimum font size of at least 16px with text scaling enabled.

High Contrast Mode: Understands the user's visual disability: minimum 4.5:1 contrast ratio

Keyboard Navigation: Full keyboard functionality for users who cannot use a mouse

2.4.2 Documentation and Development Standards

Comprehensive documentation will be maintained throughout the system development lifecycle for transparency, collaboration, and future maintenance.

Documentation Requirements:

System Documentation: Full API documentation, including Swagger/OpenAPI specification.

User Manuals: Role-specific user guides by screenshots and step-by-step instructions

Technical Documentation: Architecture diagrams, database schema, deployment guides

Code Documentation: Comments inline, following JSDoc/JavaDoc standards with a minimum of 30% comment density.

Change Logs: Detailed record of all changes, enhancements, and bug fixes

Testing Documentation: Test cases, results, and coverage reports

Compliance Documentation: Data privacy, security, and accessibility compliance reports

Tools for Documentation:

GitHub: A primary repository for code version control, issue tracking, and project management.

Confluence/Google Docs: Collaborative platform for requirements specification documentation

Draw.io/Lucidchart: To create diagrams for system architecture and workflow diagrams

SwaggerHub: API documentation and testing

Jira/ClickUp : for tracking project management and the requirement traceability matrix.

Postman: testing and documentaci3n API

2.4.3 Performance Characteristics

1. Speed & Responsiveness

The system shall perform at exceptional levels with very minimal latency; as such, the aim is to ensure that information access is fast and that updates are real-time across campuses.

Performance Indicators:

Page Load Time: ≤ 3 seconds for an initial page load, ≤ 1 second for subsequent page loads.

Notice Display: ≤ 2 seconds to load and display notice lists with filtering applied

Search Response: ≤ 1.5 seconds for search queries with up to 10,000 records

Database Queries: ≤ 100 ms for common CRUD operations

File Upload: ≤ 5 seconds for 10MB attachments on 10Mbps connections

Concurrent Users: Supporting 5,000 concurrent users at peak hours when the students are taking their exams.

API Response Time: All endpoints have 95 percent or less response time less than or equal to 200 ms.

2. Advanced Reliability and Availability

It will also ensure exceptional uptimes, deploying redundant systems to ensure this environment is continuously available for all university stakeholders.

Reliability Requirements

System Availability: 99.9% up-time, maximum of 8.76 hours of downtime annually

Scheduled Downtime: Only in non-peak hours (10:00 PM - 5:00 AM) and with 48-hour prior notice

Data Integrity: 100% data consistency, with ACID compliance for critical transactions

Success of Backup: 100% automated backup that is also verified.

Failover: Events automatically fail over to secondary servers in less than 60 seconds of primary failure

Disaster Recovery: RPO $\leq$ 15 minutes; RTO $\leq$ 4 hours

3. Scalability to Meet Growing Demand

The system architecture will support horizontal scaling and vertical scaling for higher user bases and larger volumes of data without performance degradation.

Scalability Requirements:

User Scalability: Support growth from initial 50,000 to 100,000 users without architectural changes

Notice Volume: Handle up to 10,000 new notices per month with efficient storage and retrieval.

Attachment Storage: Scale from initial 500GB to 5TB without service disruption

Concurrent Sessions: Scale to support 10,000 concurrent sessions during emergency broadcasts

Database Scaling: Support the sharding and replication of database layers for performance optimization.

Load Balancing: It automatically distributes the load on numerous servers during periods of overload.

4. Advanced Security and Privacy Measures

With comprehensive security measures in place, sensitive university data will be protected and in compliance with data protection regulations.

Requirements for Security:

Data Encryption: AES-256 encryption for data at rest and TLS 1.3 for data in transit

Authentication: Multi-factor authentication for administrative accounts, password hashing using bcrypt

Access Control: RBAC - principle of least privilege

Audit Logging: Full audit trail of all system activities; retention period: 7 years

Vulnerability Management: Automatic regular security scanning, with patches deployed within 48 hours of release

Data Privacy: Compliant with the Ethiopian Data Protection Proclamation and the university policy on privacy.

Secure APIs: API key management, rate limiting to 100 requests/minute per user

Session Security: Secure session management. Automatically time out after 30 minutes of inactivity.

2.4.4 Error Handling and Extreme Conditions

Effective error handling guarantees system robustness and allows users to respond appropriately when an error occurs.

Error Handling Strategies:

Input Validation

Client-side form validation to provide immediate feedback to the user with HTML5 validation (with custom JavaScript)

Server-side data validation of all input with extensive sanitation

Upload type, size, and content file check.

Cross-site scripting (XSS) and SQL injection by using parameterized queries and encoding output

Handling Exceptions

Exception handling that is structured with logging levels correct, viz. INFO, WARN, ERROR, FATAL

Graceful degradation when dependent services are unavailable

Circuit breaker pattern for external API calls so that failures don't cascade.

Exception classes for business logic with custom error messages are represented below:

Failure Management

Fail-safe defaults prevent unauthorized access during system failure.

Transaction rollback on failed operations so data consistency remains intact.

Queue-based processing for long-running jobs with retry mechanisms

Bulkhead pattern to isolate failures in various components of a system

Error Classification and Monitoring

Priority	Type	Resolution Time
Critical	System crash, data corruption	Immediate
High	Security breaches, failed transactions	≤ 2 hours
Medium	Performance degradation, feature failures	≤ 24 hours
Low	UI inconsistencies, minor bugs	≤ 1 week

Table 2.13: Error Classification and Monitoring

User Communication:

- Clear messages in English/Amharic.
- Suggested actions: retry, contact support.
- Error codes with correlation IDs for technical support.
- Planned downtime notifications with estimated resolution.

2.4.5 Quality Attributes

1. Availability and Uptime:

Availability: 99.9% uptime excluding scheduled maintenance

Mean Time Between Failures (MTBF): Minimum 720 hours (30 days)

Mean Time To Repair (MTTR): Maximum 1 hour for critical failures

RTO: This would be 4 hours for complete system recovery.

Recovery Point Objective (RPO): maximum data loss of 15 minutes

2. Maintainability:

Modularity of Code: Loosely coupled, highly cohesive components with clear interfaces

Code Comments: More than 30% comment coverage, with full API documentation.

Test Coverage: At least 80% unit test coverage, 70% integration test coverage

Code Review: Peer review for any code changes before merging is compulsory.

Technical Debt Management: Regular refactoring cycles with tracking metrics of debt.

3. Interoperability:

API Standards: RESTful APIs according to OpenAPI 3.0 specification.

Data Formats: Support for JSON, XML, CSV, and PDF for data exchange

Calendar Integration: Compatibility with iCal, Google Calendar, and Outlook Calendar

Single Sign-On: Both SAML 2.0 and OAuth 2.0 are supported to allow for the integration with university systems.

Mobile Platforms: iOS 12+, Android 8+, with support for progressive web app.

4. Usability:

Learnability: First-time users complete core tasks within 10 minutes.

Efficiency: For expert users, common activities should be done in a minimum number of steps, via 3-5 clicks.

Memorability: User should be able to recall 80% of task knowledge after 1 week without use.

Error Rate: The user error rate for critical tasks should be less than 5%.

Satisfaction - SUS score of at least 75

2.4.6 Security Requirements

1. Authentication and Authorization:

Password Policy: A password of at least 12 characters with complexity requirements, password expires in 90 days.

Account Lockout: 5 failed attempts within 30 minutes triggers 30-minute lockout

Session Management: Secure, random session tokens, which are automatically timed out after 30 minutes.

Role-Based Access: 5-tier role hierarchy with permission configuration.

Emergency Access: Break-glass procedures including dual authentication and full logging

2. Data Protection:

Encryption: AES-256 for data at rest, TLS 1.3+ for data in transit

Data Classification: data categories are defined as Public, Internal, Confidential, and Restricted.

Data Retention: 7 years of notices, while activity logs are kept for 1 year. Data is deleted upon request.

Data Backup: Incremental daily, full weekly, retained for 30 days

Data Integrity: Hash-based verification using SHA-256 for critical data and file upload

3. Security of Networks:

Configure the Firewall: Only permit access to the administrative interfaces using whitelisting.

DDoS Protection: The rate limiting and traffic filtering of volumetric attacks.

Intrusion Detection: real-time monitoring for suspicion activities with alerts automatically.

VPN Access: This service is necessary for the requirement of a remote administrative access to the backend systems.

Network Segmentation: Public, Private, and Administrative Zones with separate networks.

4. Security of the Application:

Input Validation: Thorough validation at client and server sides

Output Encoding: Context-aware output encoding to prevent XSS attacks

Security headers implementation: Content Security Policy, X-Frame-Options, HSTS

Dependency Scanning: Routine vulnerability scanning of third-party libraries

Security Testing: Perform a penetration test every 6 months and automated security scans on a weekly basis

2.4.7 Resource Management

Efficient use of resources will be essential for performance and cost-effectiveness. These include strategies to optimally allocate computing resources and monitor their usage within the system.

1. Computing Resources

CPU Utilization: 70% average during peak hours maximum, warning at 85%

Memory Usage: $\leq 80\%$ of the allocated memory, auto-scaling when the threshold exceeded

Storage Management: Delete temporary files automatically, compress attachments

Bandwidth Throttling: This should be employed in order to avoid congestion on the network during bulk operations.

2. Database Optimization

Query performance: All queries optimized with appropriate indexes, Query time $\leq 100\text{ms}$

Connection Pooling: This involves efficient connection management through connection reuse.

Caching Strategy: multi-level caching-Redis for sessions, and CDN for statics.

Database Maintenance: Run regular index optimization and statistics updating.

3. Application Resources

Memory Leak Avoidance: performing regular memory profiling and garbage collection optimization.

Thread Management: pool of threads that can be either limited or unlimited in number

File Handle Management: Cleaning up the resources properly in case of file upload and download

Connection Management: Graceful handling of dropped connections and timeouts

4. Monitoring and Notification

Resource Monitoring: Real-time monitoring of CPU, memory, disk usage, and network.

Alert Thresholds: Thresholds can be set and configured with escalation procedures.

Capacity Planning: This provides predictive analysis based on the use trends for resource requirements.

Cost Optimization: Automatic scaling of resources as per actual usage patterns

2.4.8 Constraints and Pseudo Requirements

1. Infrastructure Constraints :

Internet Connectivity: Single-campus, low-bandwidth $\leq 2\text{Mbps}$, offline mode for mobile, power outages handled gracefully.

2. Regulatory Constraints: Data stays in Ethiopia, comply with local privacy laws, record retention per BIT policy.

3. Organizational Constrains: Budget-limited, local IT skills, phased rollout, integrate with BIT legacy systems.

4. Technical Limitations: Browser Chrome 60+, Firefox 52+, Edge 44+, Mobile Android 8+, iOS 12+, support PDF/DOCX/XLSX/JPEG/PNG/MP4, Unicode (Amharic).

5. Environmental Limitations: Energy-efficient, reduce paper by 95%, longer hardware life, optimized server utilization.

2.5 System Requirements

2.5.1 Hardware Requirements

Server Infrastructure

Component	Minimum	Recommended
CPU	8 cores	16 cores
RAM	16GB	32GB
Storage	500GB SSD	1TB NVMe
Network	1 Gbps	10 Gbps
Backup Storage	1TB	5TB

Client Requirements

Device Type	Minimum Specifications
Desktop	Windows 10, 4GB RAM, 1280x720
Laptop	Windows 10/macOS, 4GB RAM
Tablet	iOS 12+, Android 8+, 1024x768
Mobile	iOS 12+, Android 8+, 360x640

Digital Display Requirements

Display Type	Quantity	Specifications
Campus Entry Displays	7	55" LED, 4K resolution
Department Displays	50+	43" LED, Full HD
Administrative Area	20+	32" LED, HD Ready

Table 2.14: Hardware Requirements

2.5.2 Software Requirements

Server-Side Software Stack

Component	Version	Purpose
Operating System	Ubuntu 20.04 LTS+	Server OS
Database	PostgreSQL 13+	Primary database
Cache	Redis 6.0+	Session & cache
Search	Elasticsearch 7.x	Full-text search
Web Server	Nginx 1.20+	Reverse proxy
Runtime	Node.js 18 LTS	Backend runtime

Client Stack

Component	Version	Purpose
Web Browser	Chrome 90+, Firefox 88+	Web interface
Mobile OS	Android 8+, iOS 12+	Native apps
PWA Support	Service Workers	Offline capability

Development Tools

Tool	Version	Purpose
VS Code	1.70+	Code editor
Git	2.35+	Version control
Docker	20.10+	Containerization
Postman	9+	API testing
Jest/Cypress	Latest	Testing
Swagger/OpenAPI	3.0	API documentation

Third-Party Services

Service	Provider	Purpose
Email	University SMTP/SendGrid	Email notifications
SMS	Ethio Telecom/Twilio	SMS alerts
Push Notifications	FCM/APNS	Mobile push notifications

File Storage	University NAS/AWS S3	Attachment storage
CDN	Cloudflare/Akamai	Static asset delivery
Monitoring	Prometheus+Grafana	System monitoring

Compatibility Requirements

Area	Requirement	Area
File Formats	PDF, DOCX, XLSX, JPEG, PNG, MP4	File Formats
Export Formats	CSV, Excel, PDF, JSON	Export Formats
Calendar Integration	iCal, Google Calendar, Outlook	Calendar Integration
Single Sign-On	SAML 2.0, OAuth 2.0	Single Sign-On
Accessibility	NVDA, JAWS, VoiceOver compatible	Accessibility
Text Scaling	200% zoom without loss	Text Scaling

Table 2.15: Software Requirements

2.5.3 Network Requirements

Minimum Specifications

Requirement	Specification
Latency	≤100ms round-trip (campus)
Bandwidth	10Mbps per 100 concurrent users
Uptime	99.5% network availability
Security	TLS 1.2+ for external connections

Campus Network Requirements

- **Internal Network:** Gigabit Ethernet between buildings
- **Wi-Fi Coverage:** Complete academic/administrative areas

- **Mobile Data:** 3G/4G coverage for mobile access
- **Load Balancing:** Distributed access across campuses
- **Security:** Network segmentation, intrusion detection

2.5.4 Backup and Recovery Requirements

Backup Schedule

Backup Type	Schedule	Retention
Incremental	Every 4 hours (operational)	7 days
Full Daily	2:00 AM daily	30 days
Weekly	Sunday 2:00 AM	12 months
Monthly	First of month	7 years

Recovery Objectives

Objective	Target	Measurement
RTO (Recovery Time)	4 hours	System recovery time
RPO (Recovery Point)	15 minutes	Maximum data loss
Testing Frequency	Quarterly	Disaster recovery drills

Storage Requirements

Storage Type	Capacity	Purpose
Online Storage	1TB	Active data
Nearline Storage	2TB	Recent backups (30-day)
Offline/Archival	5TB	Long-term archives (7-year)

Table 2.16: Network Requirements

2.6 System Design

2.6.1 System Architecture Diagram

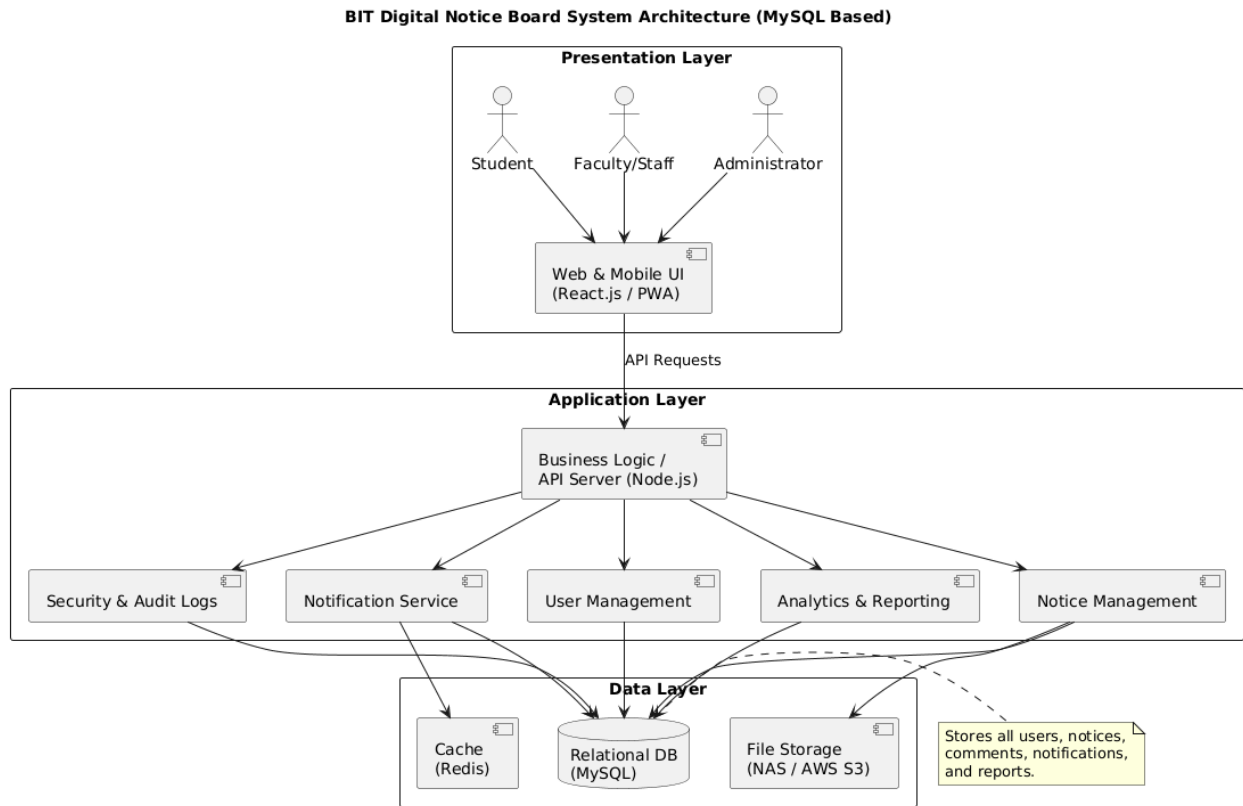


Figure 2.17: System Architecture Diagram

2.6.2: Conceptual modeling: Class diagram

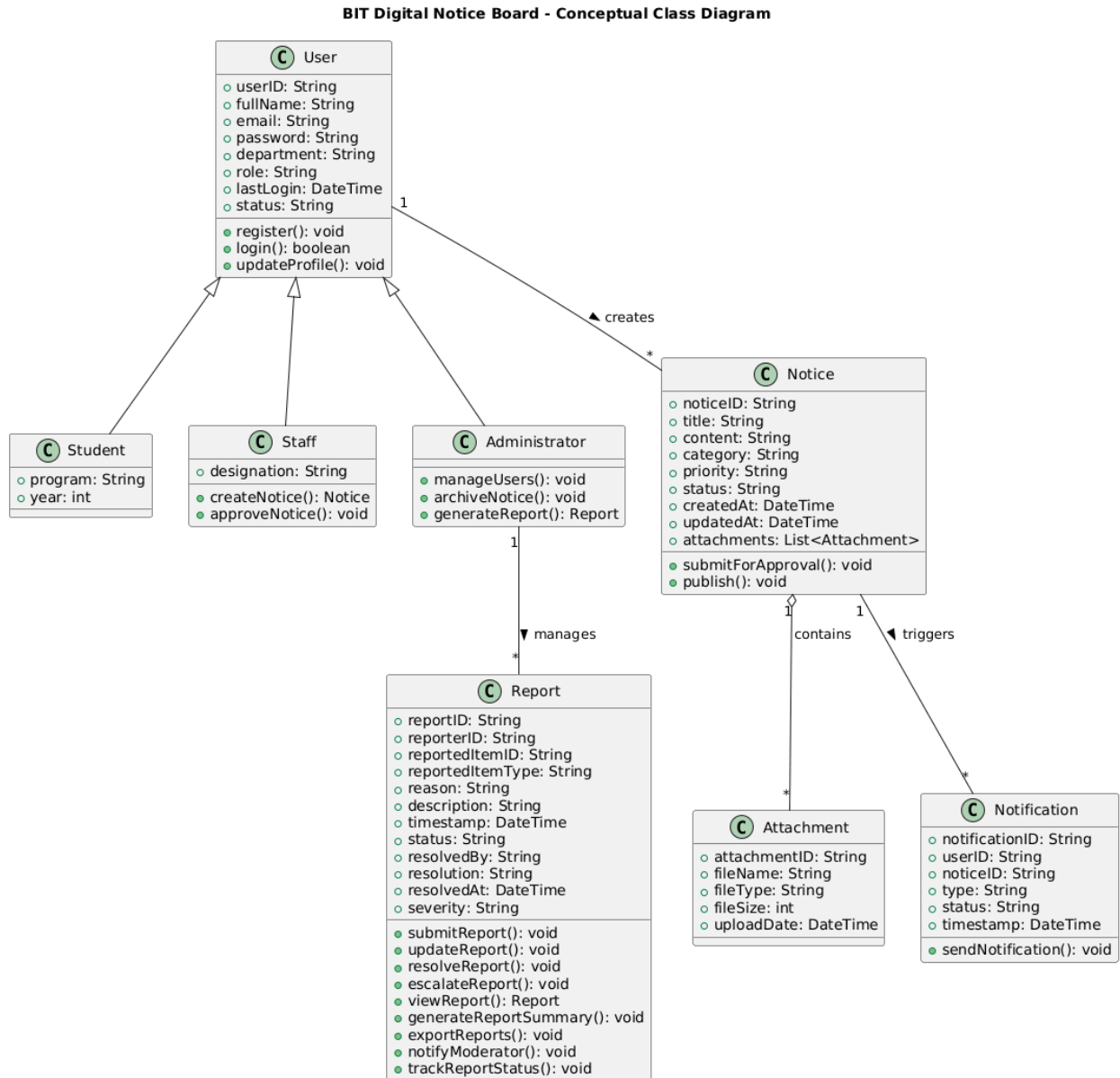


Figure 2.18: Class diagram

2.6.3: Identifying Change Cases

Change Case 1: AI-Powered Content Creation and Recommendation

- **Change:** Implementing artificial intelligence algorithms for personalized notice recommendations, automated content categorization, and intelligent spam/duplicate detection.
- **Likelihood: High.** AI integration in content management systems is increasingly common, with growing demand for personalized digital experiences in educational and organizational settings.
- **Impact:**
 - **High:** Enhanced user experience through personalized content delivery, improved content organization, and automated moderation of inappropriate content.
 - **Moderate:** Increased system complexity and computational requirements, potential privacy concerns with user data analysis.
 - **Low:** Risk of algorithmic bias in content recommendation, requiring careful training and validation of AI models.

Change Case 2: Mobile Push Notifications and Real-time Updates

- **Change:** Expanding notification delivery from traditional email to mobile push notifications, SMS alerts, and real-time dashboard updates with WebSocket technology.
- **Likelihood: High.** Mobile device penetration continues to grow, and users increasingly expect instant, multi-channel notifications for important information.
- **Impact:**
 - **High:** Improved information dissemination speed, increased user engagement, better accessibility for mobile-centric users.
 - **Moderate:** Development and maintenance costs for multiple notification channels, potential notification fatigue if not properly managed.

- **Low:** Platform-specific development requirements (iOS, Android), battery consumption concerns for mobile devices.

Change Case 3: Integration with External Calendar and Productivity Tools

- **Change:** Adding API integration with popular calendar applications (Google Calendar, Outlook) and productivity tools (Microsoft Teams, Slack) for seamless notice synchronization.
- **Likelihood: Medium-High.** Modern digital ecosystems increasingly favor interconnected applications, with users expecting seamless workflow integration.
- **Impact:**
 - **High:** Enhanced user convenience, improved adoption rates, better integration with existing organizational workflows.
 - **Moderate:** Development complexity for multiple API integrations, ongoing maintenance for third-party API changes.
 - **Low:** Data synchronization conflicts, potential security concerns with external API access.

Change Case 4: Advanced Analytics and Reporting Dashboard

- **Change:** Implementing comprehensive analytics features including notice view tracking, user engagement metrics, content performance analysis, and automated reporting.
- **Likelihood: Medium.** While valuable, advanced analytics represent a secondary feature that may follow core system stabilization.
- **Impact:**
 - **High:** Data-driven decision making for administrators, improved content strategy, measurable ROI demonstration.
 - **Moderate:** Increased database requirements for analytics storage, potential performance impact on high-traffic systems.

- **Low:** Privacy considerations for detailed user tracking, training requirements for interpreting analytics data.

Change Case 5: Multi-language and Accessibility Support

- **Change:** Adding support for multiple languages (translation features) and enhanced accessibility features (screen reader compatibility, keyboard navigation, high contrast modes).
- **Likelihood: Medium.** Essential for global or diverse user bases but may be prioritized based on specific institutional requirements.
- **Impact:**
 - **High:** Expanded user base, compliance with accessibility regulations (ADA, WCAG), improved user experience for diverse populations.
 - **Moderate:** Translation and localization costs, ongoing maintenance for multiple language versions.
 - **Low:** Interface layout challenges with different language text lengths, testing complexity for various accessibility scenarios.

Change Case 6: Blockchain-Based Notice Authentication

- **Change:** Utilizing blockchain technology for tamper-proof notice verification, immutable audit trails, and secure timestamping of critical announcements.
- **Likelihood: Low-Medium.** While offering significant security benefits, blockchain integration represents a substantial architectural change with current adoption barriers.
- **Impact:**
 - **High:** Enhanced security and trust in official notices, verifiable notice authenticity, protection against content tampering.
 - **Moderate:** Substantial development and infrastructure costs, increased system complexity, specialized expertise requirements.

- **Low:** Energy consumption concerns with certain blockchain implementations, integration challenges with existing systems.

Change Case 7: Single-Campus Adaptation for BIT

- **Change:** Adapt the Digital Notice Board System from a multi-campus BDU setup to a single-campus BIT setup. All notices, users, notifications, and displays will be confined to BIT campus only.
- **Likelihood:** High. Since BIT has only one campus, this adaptation is necessary for proper deployment.
- **Impact:**
 - **High:** Simplifies system architecture, notice distribution, and network management. Reduces complexity in user roles and display management.
 - **Moderate:** Minor adjustments in notification targeting, display configuration, and reporting modules.
 - **Low:** Minimal effect on existing core functionalities; mostly configuration and interface updates.

2.6.4:User Interface Prototyping

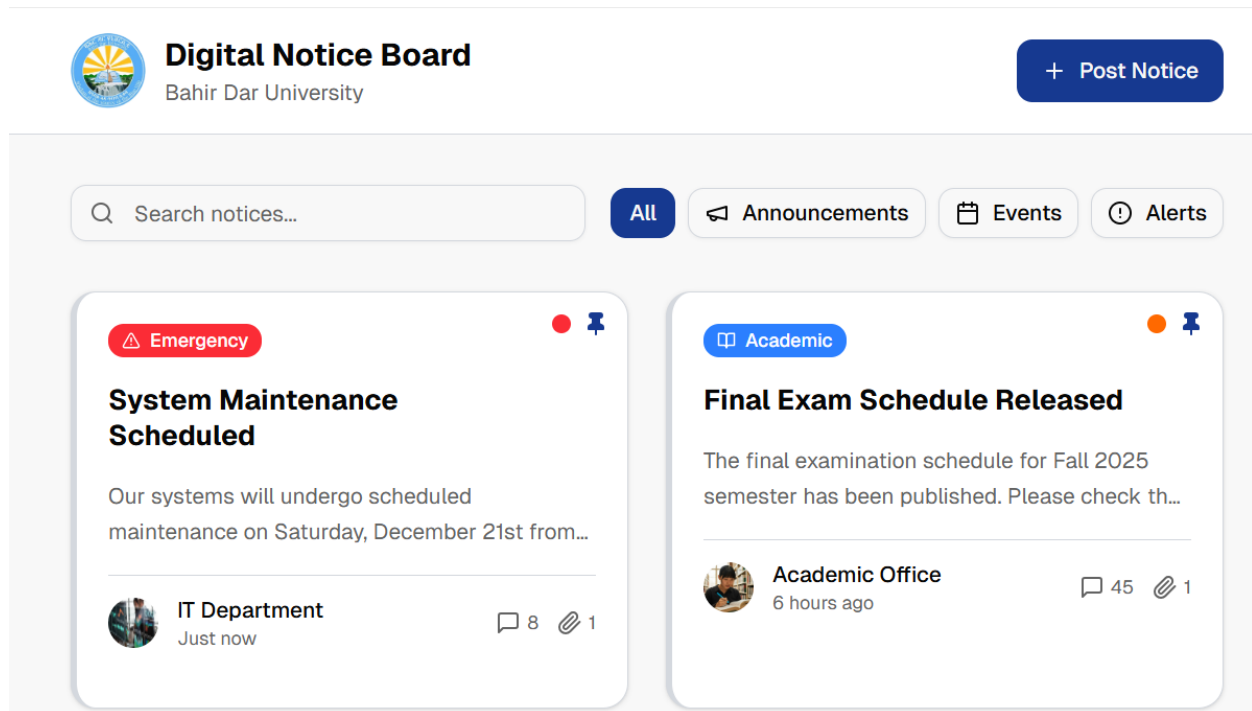


Figure 2.19: Dashboard user interface



Digital
Bahir Dar University

Emergency

System Maintenance Scheduled

Our systems will be down for maintenance on December 15th from 10:00 AM to 12:00 PM.

 IT Department
Just now

5

Post Notice

Alerts

×

Create New Notice

Fill out the form below to post a new notice to the board

Title *

0/100

Enter a clear and concise title

Category *

Select a category

Priority Level *

☐ Low ☒ Medium ☐ High ☐ Urgent

Content *

Write your notice content here. Be clear and detailed.

Content *

Write your notice content here. Be clear and detailed.

Attachments

Drag and drop files here, or click to browse

Choose Files

Visibility *

- ☒ Public - Everyone can see
- ☐ Department - Specific department only

Schedule Publishing (Optional)

mm/dd/yyyy

Visibility *

- ☒ Public - Everyone can see
- ☐ Department - Specific department only
- ☐ Private - Restricted access

Schedule Publishing (Optional)

mm/dd/yyyy

--:-- --

Leave empty to publish immediately

Author / Department *

Your name or department name

Notifications

- ☐ Send email notifications to subscribers
- ☐ Send push notifications to mobile devices

Preview Cancel Post Notice

Figure 2.20: Create Notice user interface

Chapter 3: System Design

This chapter presents the system design of the **BIT Digital Notice Board System**, transforming the requirements defined in the previous chapter (Requirements Specification) into a detailed technical blueprint for system development. The system design phase focuses on both high-level and low-level design decisions, including system architecture, data organization, user interfaces, subsystems, and security mechanisms.

The Digital Notice Board System is designed to serve **BIT**, providing a centralized platform for publishing, managing, and accessing official notices. The design emphasizes reliability, scalability, security, and ease of use to ensure effective communication between administrators, faculty/staff, and students. This chapter serves as a guideline for developers and stakeholders, ensuring the system is implemented consistently with institutional and technical requirements.

Design Goals

Design goals describe the quality attributes the system must optimize to satisfy non-functional requirements (NFRs). These goals guide design decisions to ensure that the Digital Notice Board System is functional, efficient, secure, scalable, and maintainable.

The major quality perspectives considered include:

- Functionality
- Performance
- Dependability
- Maintainability
- Scalability

Functionality

The system is designed to fully satisfy the specified functional requirements by providing comprehensive features for notice management and dissemination. The key functional objectives include:

- Allow administrators to create, update, approve, and delete notices.
- Enable faculty and staff to publish authorized notices.
- Allow students to view, search, and receive notifications for notices.
- Support categorization, prioritization, and archiving of notices.
- Provide real-time notifications and access to historical notices.

Performance

High performance is essential for the Digital Notice Board System due to its role in disseminating time-sensitive information. The system is designed to:

- Provide fast response times for notice retrieval and search operations.
- Support concurrent access by users across multiple campuses.
- Efficiently handle frequent updates and notifications without degradation.

Optimized database queries, caching mechanisms, and lightweight client-server communication are used to ensure smooth system performance.

Dependability

1. Reliability

The system ensures reliable operation through stable hosting infrastructure and consistent monitoring. Notices remain accessible with minimal downtime, ensuring uninterrupted communication.

2. Availability

The system is available 24/7 and supports access from different campuses and devices through web-based deployment.

3. Security

Security is enforced through:

- Secure authentication and authorization mechanisms.

- Role-based access control for administrators, staff, and students.
- Data encryption during transmission.
- Audit logs for sensitive operations.

4. Error Handling

The system integrates robust error-handling mechanisms, including:

- Input validation to prevent incorrect data entry.
- Exception handling and logging for system errors.
- User-friendly error messages to guide users during failures.

Maintainability

1. Documentation

Comprehensive system documentation is maintained to support future updates and enhancements.

2. Code Readability

Modular design and clear coding standards improve readability and ease of maintenance.

3. Error Monitoring

Continuous monitoring and logging help detect and resolve issues promptly.

4. Scalability

The system is designed to scale horizontally, allowing additional campuses and users to be added without significant redesign.

End-User Criteria

1. User-Friendly Interface

The interface is intuitive and accessible for users with varying technical skills.

2. Efficiency

Advanced search and filtering options allow users to quickly locate relevant notices.

3. **Transparency**

Notices display clear metadata such as publisher, date, category, and priority.

4. **Convenience**

Users can access notices anytime and from any location using web-enabled devices.

3.1 **Architectural Design**

The Digital Notice Board System follows a **three-tier architecture model**, improving modularity, security, and scalability.

1. **Presentation Layer**

- Provides the user interface for students, staff, and administrators.
- Accessible through web browsers.
- Communicates with the application layer via HTTP/HTTPS.

2. **Application Layer**

- Handles business logic, notice validation, and access control.
- Acts as a bridge between the presentation and data layers.
- Enforces system rules and workflows.

3. **Data Layer**

- Manages data storage and retrieval.
- Stores user data, notices, logs, and notifications.
- Ensures data integrity and consistency.

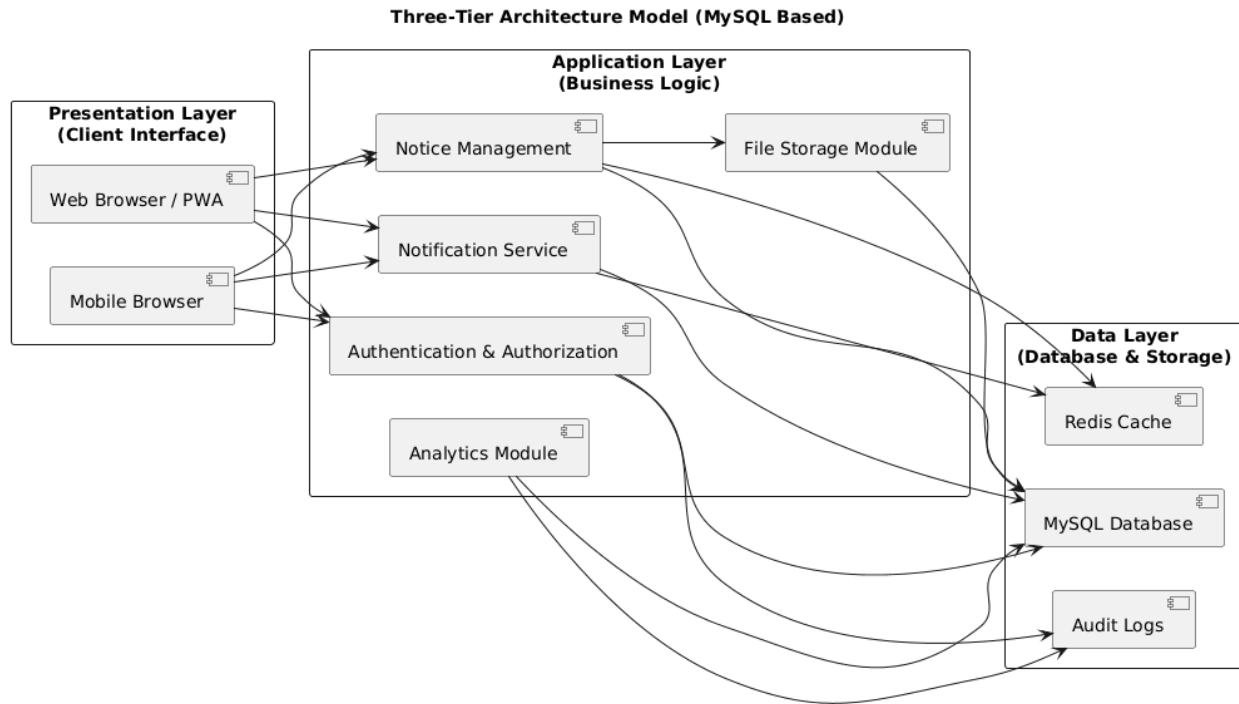


Figure 3.1: Three-tier architecture model

Subsystem Decomposition

The system is decomposed into the following subsystems:

- User Interface Subsystem
- User Management Subsystem
- Notice Management Subsystem
- Notification Subsystem
- Security and Access Control Subsystem
- Data Management Subsystem

Each subsystem operates independently but communicates through defined interfaces to ensure smooth integration.

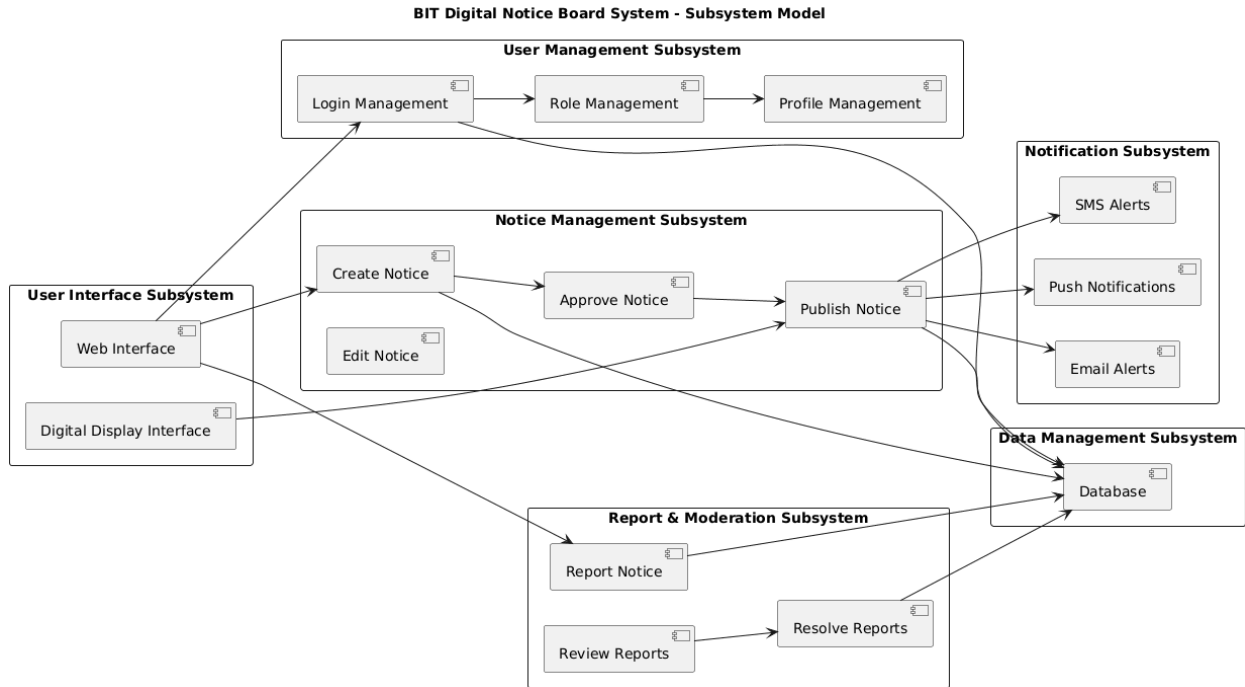


Figure 3.2: Subsystem model

3.1.1 Component Modeling

The Digital Notice Board System adopts a component-based architecture. Each component encapsulates specific responsibilities and interacts with others through well-defined interfaces.

Major components include:

- User Interface Component
- Authentication and Authorization Component
- Notice Management Component
- Notification Component
- Reporting and Logging Component
- Database Management Component

This modular structure improves maintainability, flexibility, and system evolution.

Figure 3.2: Component Diagram - BIT Digital Notice Board System

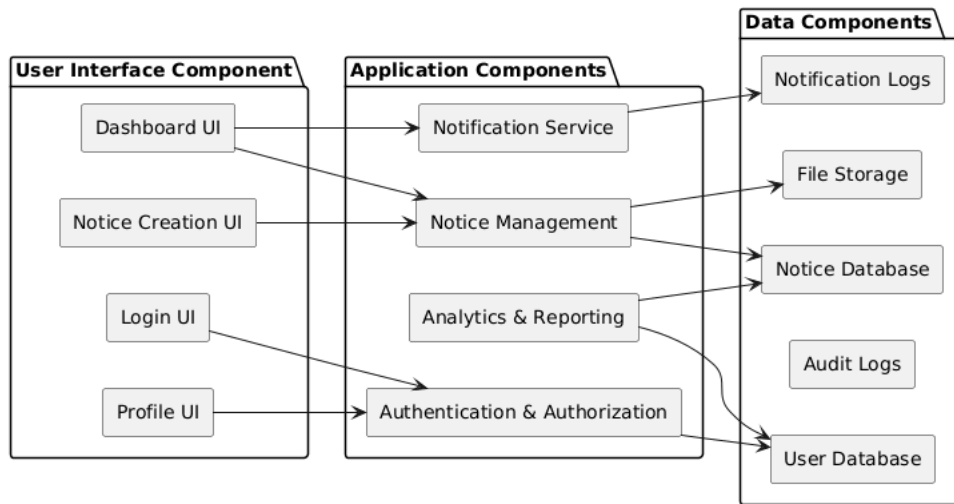


Figure 3.3: Component Diagram

3.1.2 Deployment Modeling

The system is deployed using a multi-tier web-based deployment model.

Client Layer

- Users access the system via web browsers.
- Technologies: HTML, CSS, JavaScript, React/Next.js.

Application Server

- Hosts application logic and APIs.
- Handles authentication, notice processing, and notifications.

Database Server

- Stores system data securely.
- Manages backups and recovery processes.

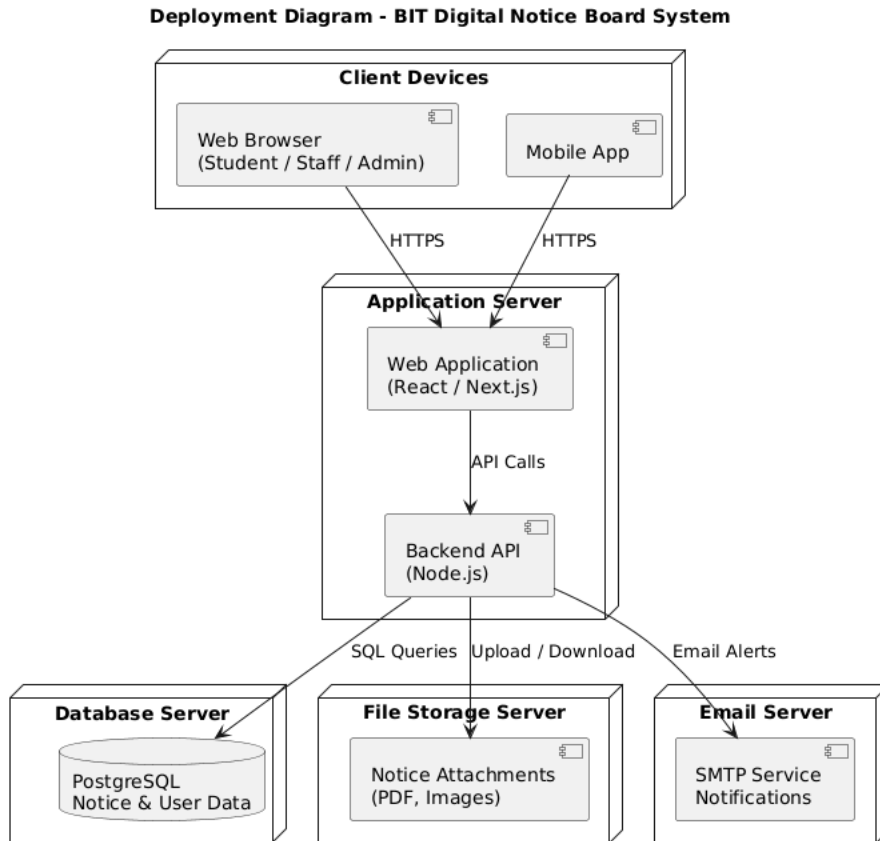


Figure 3.4: Deployment diagram

Communication Flow

- Users send requests from client devices.
- Requests are processed by the application server.
- Data is retrieved or stored in the database server.
- Responses are returned to the client.

Justification for Architecture

The chosen architecture provides:

- **Modularity:** Clear separation of concerns.

- **Scalability:** Easy expansion to additional campuses.
- **Security:** Restricted database access.
- **Performance:** Optimized load distribution.

3.2 Detail Design

3.2.1 Design Class Model

The design class model defines system classes, attributes, methods, and relationships. It supports object-oriented development and ensures clarity in system behavior.

Key classes include:

- User
- Student
- Staff
- Administrator
- Notice
- Notification
- Category
- AuditLog

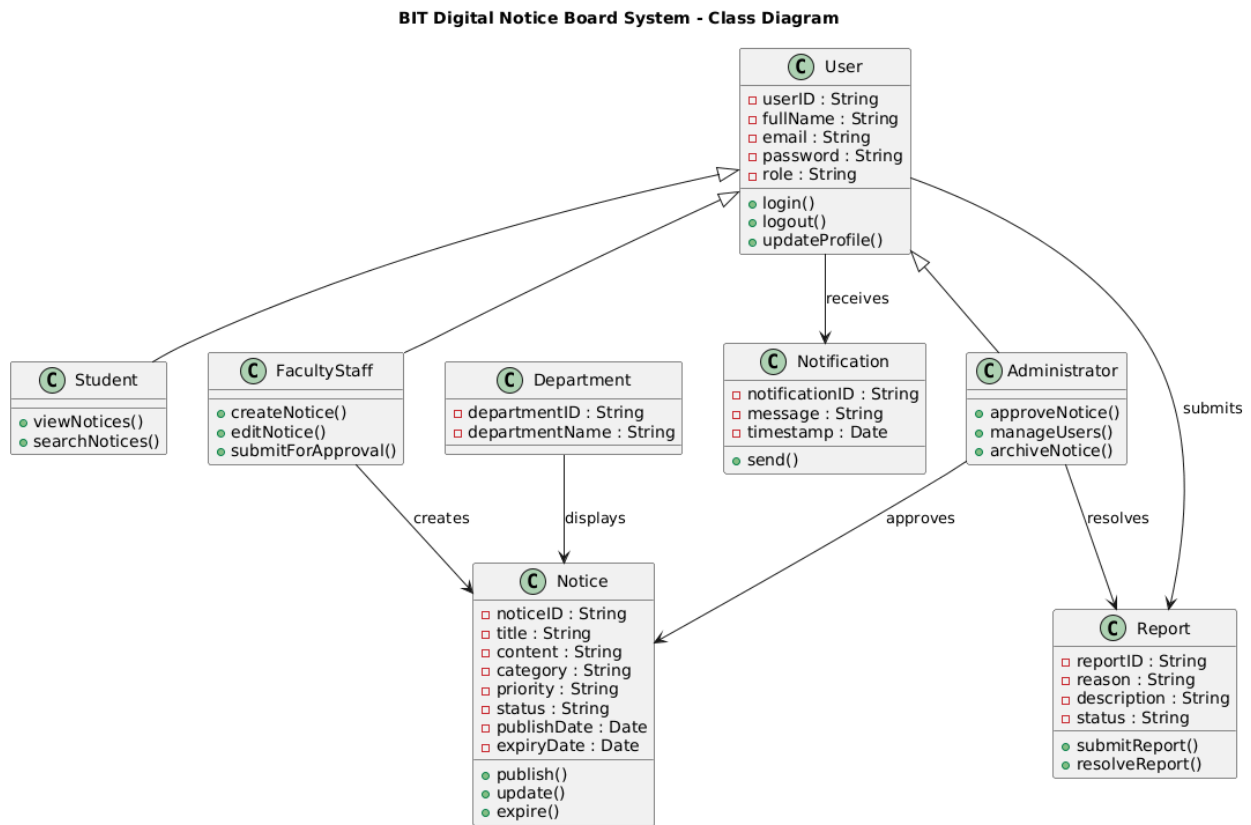


Figure 3.5: Class Diagram

Table 3.1: User entity description

Attribute Name	Data Type	Description
userID	String	Unique identifier for each user
fullName	String	Full name of the user
email	String	User email address
password	String	Encrypted login password
role	String	User role (Student, Faculty/Staff, Admin)

Table 3.2: Student entity description

Attribute Name	Data Type	Description
studentID	String	Unique student identifier
department	String	Student's department
year	Integer	Academic year of study

Table 3.3: Faculty/Staff entity description

Attribute Name	Data Type	Description
staffID	String	Unique staff identifier
department	String	Department of the staff member
position	String	Academic or administrative position

Table 3.4: Adminstrator entity description

Attribute Name	Data Type	Description
adminID	String	Unique administrator identifier
office	String	Administrative office

Table 3.5: Notice entity description

Attribute Name	Data Type	Description
noticeID	String	Unique notice identifier
title	String	Notice title
content	Text	Detailed notice content
category	String	Type of notice (Academic, Event, Exam, etc.)
priority	String	Notice priority level

status	String	Notice status (Pending, Approved, Published)
publishDate	Date	Date the notice is published
expiryDate	Date	Notice expiration date

Table 3.6: Department entity description

Attribute Name	Data Type	Description
departmentID	String	Unique department identifier
departmentName	String	Name of the department

Table 3.7: Notification entity description

Attribute Name	Data Type	Description
notificationID	String	Unique notification identifier
message	String	Notification message
timestamp	DateTime	Time notification was sent

Table 3.8: Report entity description

Attribute Name	Data Type	Description
reportID	String	Unique report identifier
reason	String	Reason for reporting
description	Text	Detailed explanation
status	String	Report status (Open, Resolved)

3.2.2 Entity-Relationship (E-R) Model

The E-R model depicts relationships between **users**, **notices**, **comments**, **notifications**, and **reports**.

- **User:** Primary entity.
- **Notice:** Created by a user, associated with departments.
- **Comment:** Linked to notice and user.
- **Notification:** Linked to notices and users.
- **Report:** Submitted by users against notices or comments.

BIT Digital Notice Board System - E-R Diagram

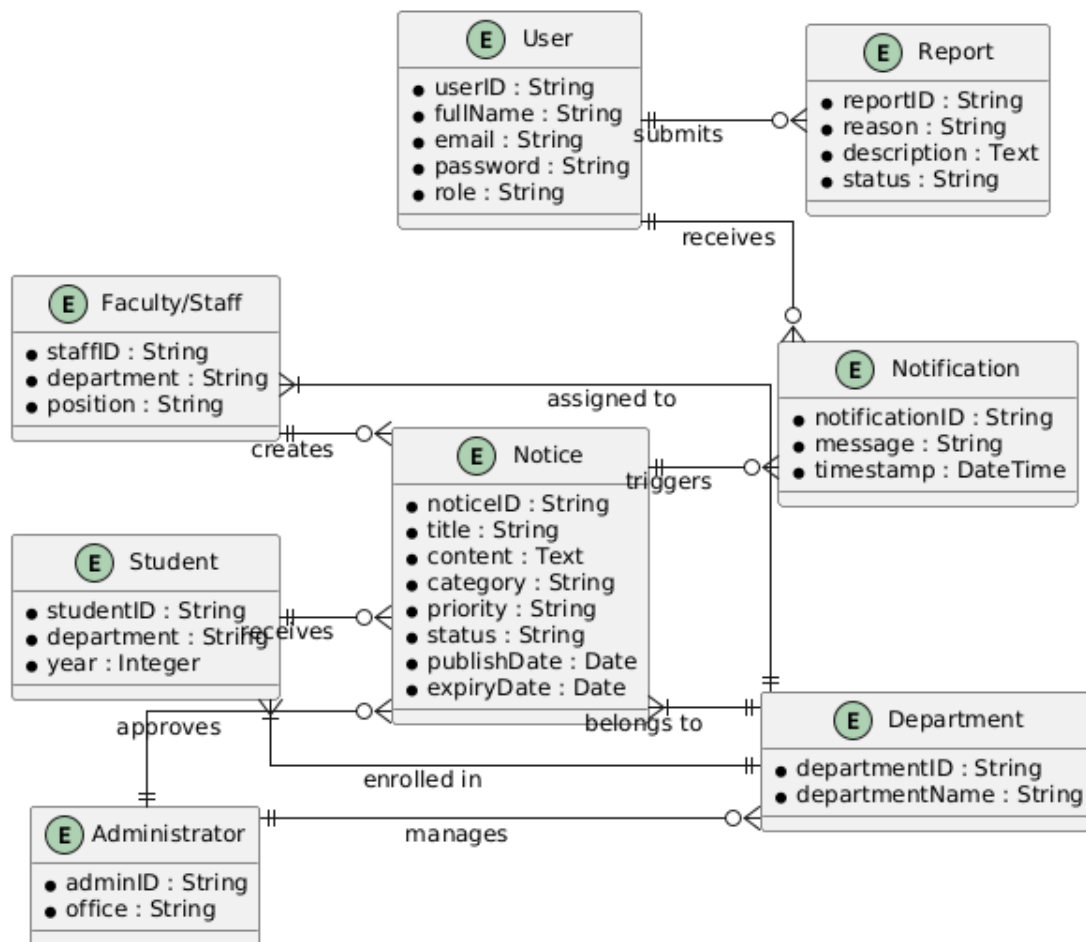


Figure 3.6: E-R Diagram

3.3 User Interface Design

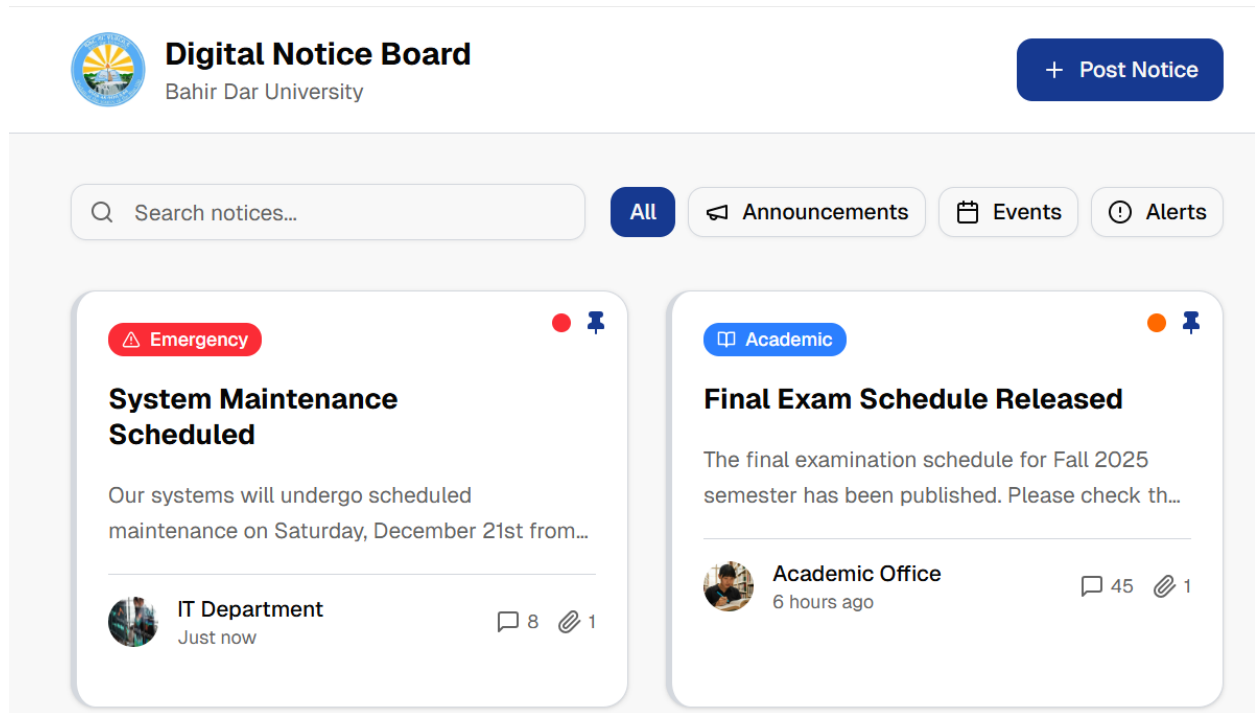


Figure 3.7: Dashboard UI

3.4 Access Control and Security

The **BIT Digital Notice Board System** implements strong access control and security mechanisms to ensure that different users have appropriate permissions based on their assigned roles. The system enforces role-based access control (RBAC) to protect sensitive information, prevent unauthorized operations, and maintain data integrity.

Users of the system are categorized into three main roles:

- **Administrator**
- **Faculty/Staff**
- **Student**

Each role has clearly defined permissions that control what actions the user can perform within the system. Security mechanisms such as authentication, authorization, encrypted communication, and audit logging are applied to ensure safe and reliable system operation.

Tasks / Functions	Administrator	Faculty/Staff	Student
Register and Login	✓	✓	✓
View Notices	✓	✓	✓
Search and Filter Notices	✓	✓	✓
Create Notice	✓	✓	✗
Edit Own Notice	✓	✓	✗
Delete Notice	✓	✗	✗
Approve Notices	✓	✗	✗
Schedule Notice Publication	✓	✓	✗
Upload Attachments	✓	✓	✗
Receive Notifications	✓	✓	✓
Manage Users and Roles	✓	✗	✗
Archive Expired Notices	✓	✗	✗
View Analytics and Reports	✓	✗	✗
Update Profile	✓	✓	✓
Access Audit Logs	✓	✗	✗

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Appendices

A. Current Physical Notice Board

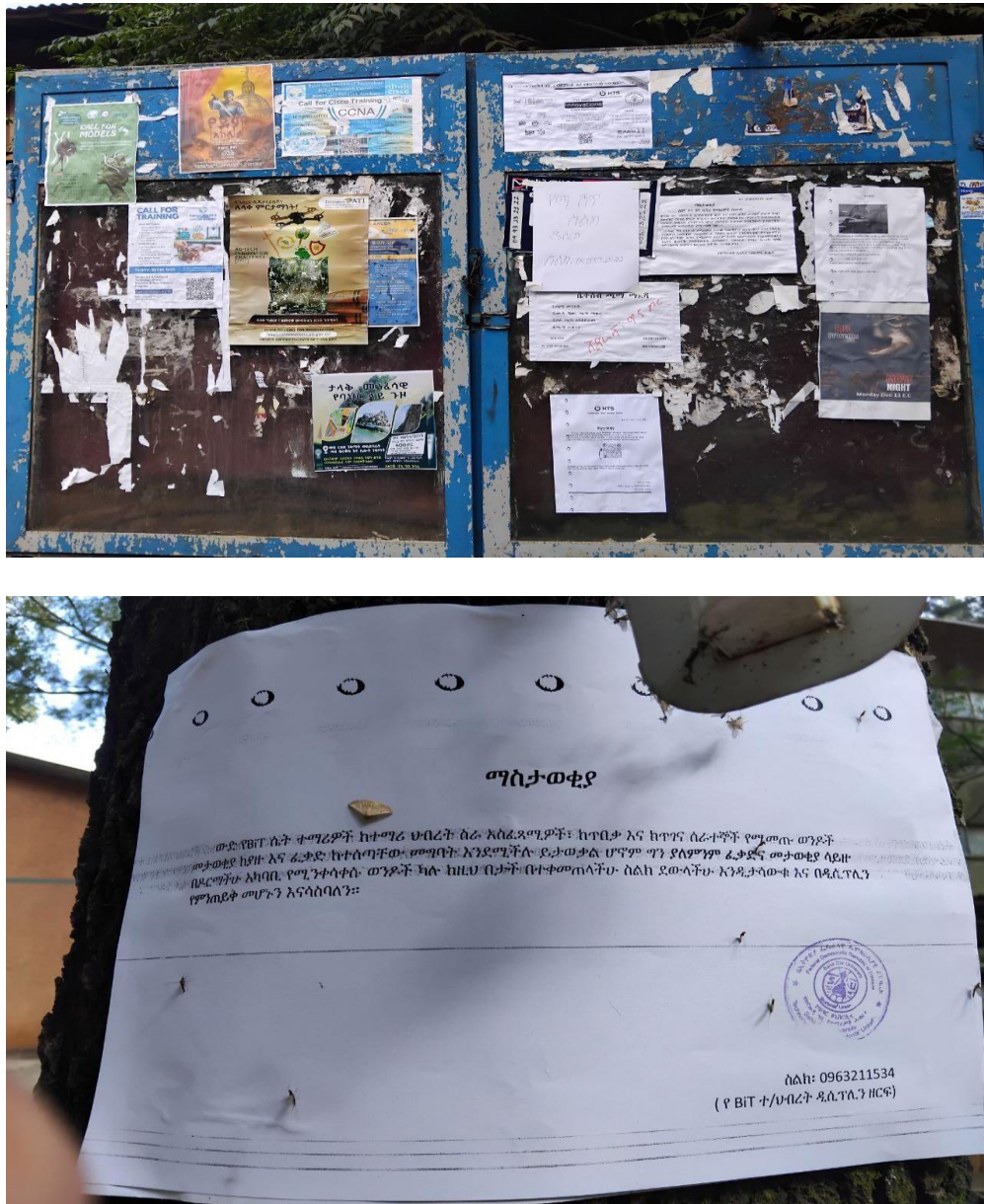


Figure 3.8: Current Physical Notice Board Diagram

B. Proposed Digital Notice

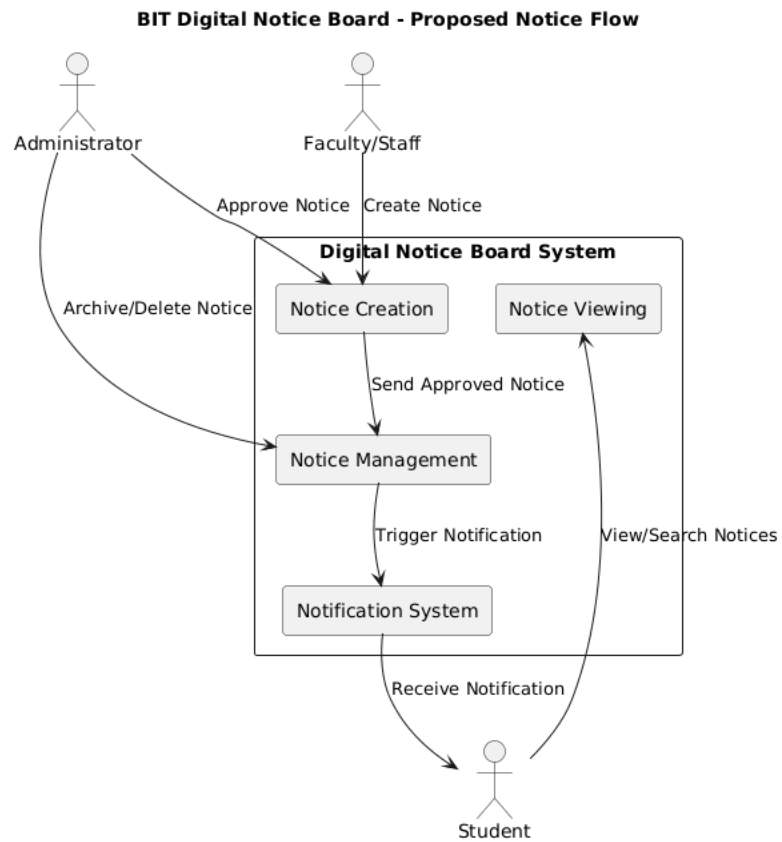


Figure 3.9: Proposed Digital Notice Simple Flow